Inbroduction

```
!pip install factor_analyzer
```

```
Requirement already satisfied: factor_analyzer in /usr/local/lib/python3.12/dist-packages (0.5.1)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from factor_analyzer) (2.2.2)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from factor_analyzer) (1.16.2)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from factor_analyzer) (2.0.2)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (from factor_analyzer) (1.6.1)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->factor_analyzer) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->factor_analyzer) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->factor_analyzer) (2025.2)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn->factor_analyzer) (1.5.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn->factor_analyzer) (3.6.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->factor_analyzer) (1.17.0)
```

```
import pandas as pd
import numpy as np
from google.colab import files
import datetime as dt
import seaborn as sns
import statsmodels.api as sm
import matplotlib.pyplot as plt
import scipy.stats as stats
from scipy.stats import shapiro, jarque_bera
from matplotlib.patches import Ellipse, Arc
import sympy as sp
import datetime as dt
import pandas_datareader.data as web
from sklearn.decomposition import PCA
from factor_analyzer import FactorAnalyzer
from scipy.stats import lognorm
import math
import random
```

```
df = pd.read_csv("/content/drive/MyDrive/Quant/sp500.csv")
```

```
df.head()
```

	Date	Open	High	Low	Close
0	1975-09-15	83.300003	83.489998	82.290001	82.879997
1	1975-09-16	82.879997	83.430000	81.790001	82.089996
2	1975-09-17	82.089996	82.930000	81.570000	82.370003
3	1975-09-18	82.370003	84.339996	82.230003	84.059998
4	1975-09-19	84.260002	86.389999	84.260002	85.879997

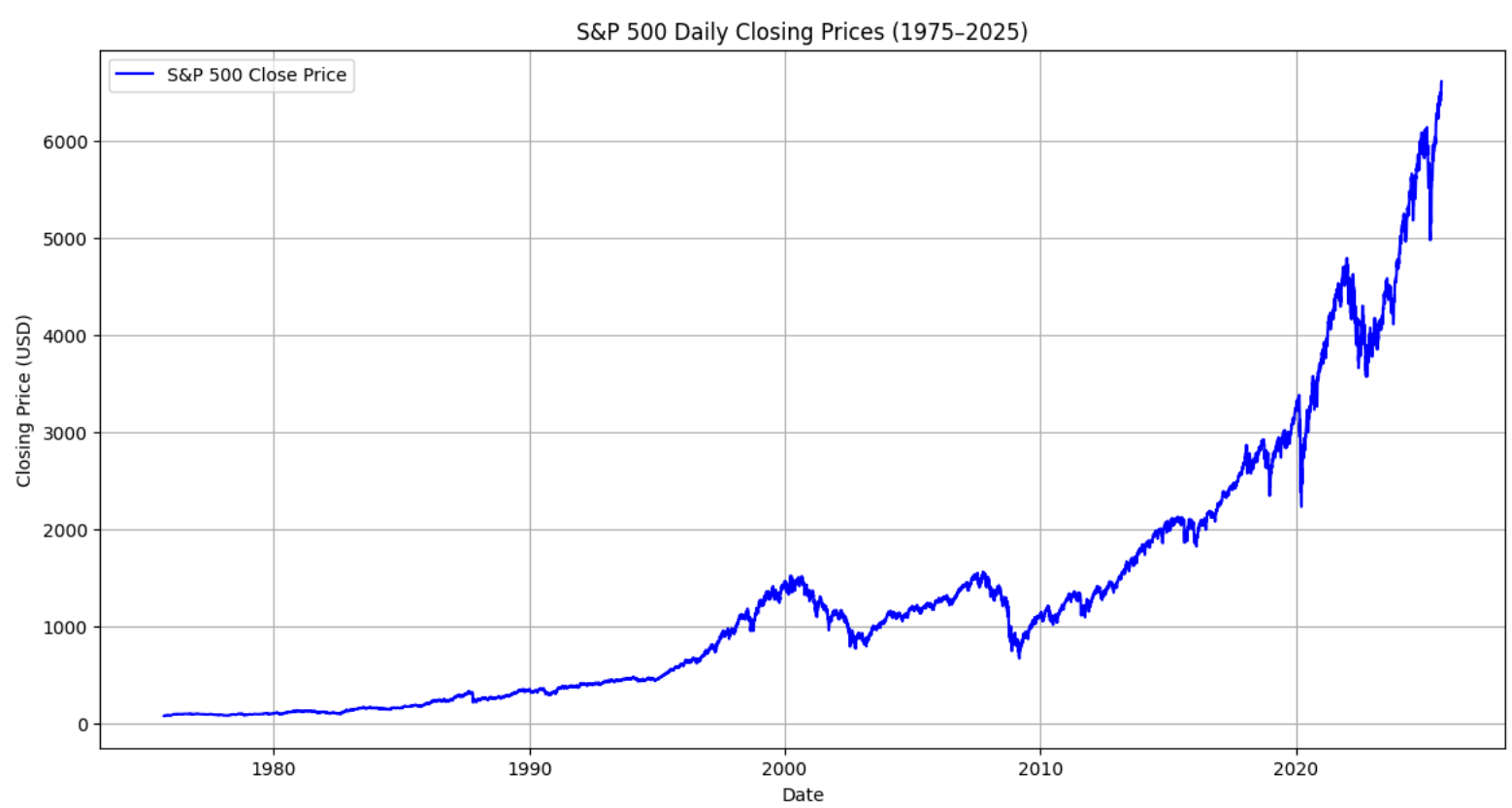
Next steps: Generate code with df New interactive sheet

```
# Make sure index is datetime
df.index = pd.to_datetime(df.index)
df['Date'] = pd.to_datetime(df['Date'])
df.set_index('Date', inplace=True)
```

```
# Filter the period
df_filtered = df.loc["1975-09-15":"2025-09-15"]
```

Ex. 1 (a)

```
# Plot Close price
plt.figure(figsize=(14,7))
plt.plot(df_filtered.index, df_filtered['Close'], color="blue", label="S&P 500 Close Price")
plt.title("S&P 500 Daily Closing Prices (1975–2025)")
plt.xlabel("Date")
plt.ylabel("Closing Price (USD)")
plt.legend()
plt.grid(True)
plt.show()
```



Ex. 1 (b)

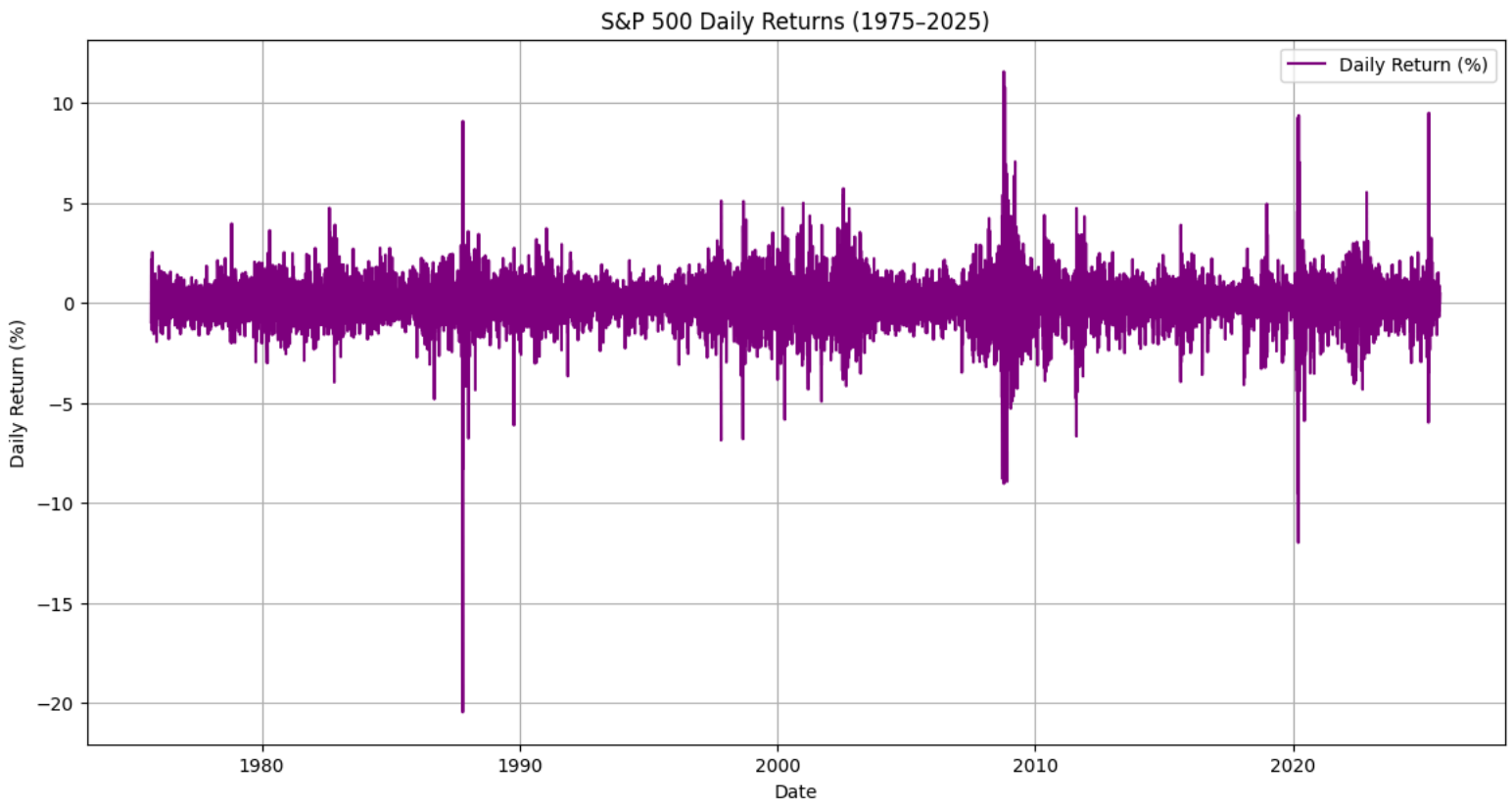
The daily return is calculated as:

$$R_t = (\frac{P_t - P_{t-1}}{P_{t-1}}) \times 100 = (\frac{P_t}{P_{t-1}} - 1) \times 100$$

- R_t = daily return at time t

- P_t = closing price today
- P_{t-1} = closing price yesterday

```
# Daily return = percentage change of Close
df['Daily Return'] = df['Close'].pct_change() * 100   # in %
plt.figure(figsize=(14,7))
plt.plot(df.index, df['Daily Return'], color="purple", label="Daily Return (%)")
plt.title("S&P 500 Daily Returns (1975–2025)")
plt.xlabel("Date")
plt.ylabel("Daily Return (%)")
plt.legend()
plt.grid(True)
plt.show()
```

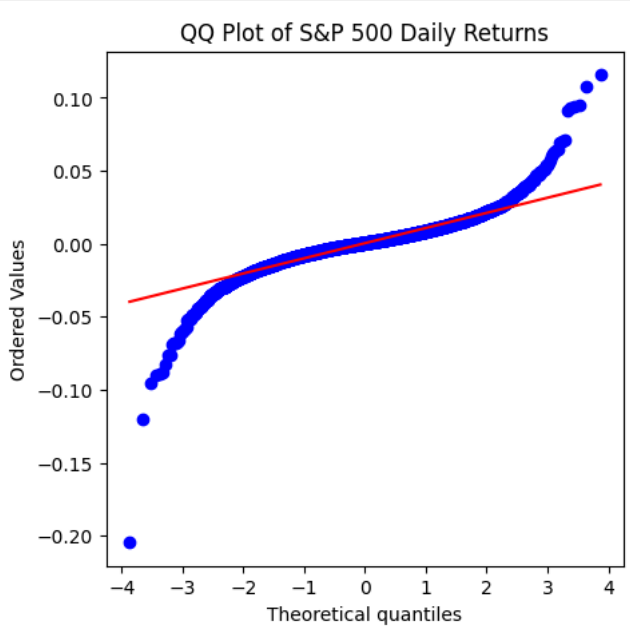


Ex. 1 (c)

> **As a rule of thumb**, the more that the points in a Q-Q plot lie on a straight diagonal line, the more normally distributed the data. **Conversely**, the more that the points deviate from a straight diagonal line, the less normally distributed the data.

The S&P 500 daily returns do not follow a normal distribution. As points deviate strongly from the line in the tails. This means there are more extreme returns (fat tails) than a normal distribution would predict.

```
# Calculate daily returns
df['Daily Return'] = df['Close'].pct_change()
returns = df['Daily Return'].dropna()

# QQ Plot
plt.figure(figsize=(5,5))
stats.probplot(returns, dist="norm", plot=plt)
plt.title("QQ Plot of S&P 500 Daily Returns")
plt.show()
```



> To formally test the normality of the S&P 500 daily return series, we can apply two statistical tests:

Shapiro–Wilk Test and **Jarque–Bera Test**.

Shapiro–Wilk Test

The Shapiro–Wilk test evaluates the null hypothesis that a sample $(x_1, x_2, \ldots, x_n)$ came from a normally distributed population. For more details: [Shapiro–Wilk test]

The test statistic is defined as:

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n} (x_i - \overline{x})^2}$$

where:

- $x_{(i)}$ are the **ordered sample values** (from smallest to largest),
- $\overline{x}$ is the **sample mean**,
- a_i are constants derived from the expected values of the order statistics of a normally distributed sample.

If the p-value associated with W is less than 0.05, we reject the null hypothesis and conclude that the data is **not normally distributed**.

Results:

- Test statistic = **0.8951544995780921**
- p-value ≈ **5.018566335926154e-68**

- Since the p-value is far **smaller than 0.05**, we reject the null hypothesis H_0 that the returns are normally distributed.
- **Interpretation:** the return series deviates significantly from a normal distribution.

```
# Shapiro-Wilk Test
shapiro_test = shapiro(returns)
print("Shapiro-Wilk Test:")
print("Statistic =", shapiro_test.statistic, ", p-value =", shapiro_test.pvalue)
```

```
Shapiro-Wilk Test:
Statistic = 0.8951544995780921 , p-value = 5.018566335926154e-68
/usr/local/lib/python3.12/dist-packages/scipy/stats/_axis_nan_policy.py:579: UserWarning: scipy.stats.shapiro: For N > 5000, computed p-value may not be accurate. Current N is 12
  res = hypotest_fun_out(*samples, **kwds)
```

Jarque–Bera Test In statistics, the **Jarque–Bera test** is a goodness-of-fit test of whether sample data have the skewness and kurtosis matching a normal distribution. The test statistic is always nonnegative. If it is far from zero, it signals the data does not have a normal distribution. For more details: [Jarque–Bera test]

The test statistic is defined as:

$$JB = \frac{n}{6}\left(S^2 + \frac{1}{4}(K-3)^2\right)$$

Where:

- n = sample size
- S = sample skewness
- K = sample kurtosis

For a perfectly normal distribution:

- S = 0 (no skewness)
- K = 3 (mesokurtic distribution)
  If the **p-value smaller than 0.05**, we reject the null hypothesis H_0 and conclude that the data is **not normally distributed**.

Results (Jarque–Bera Test):

- Test statistic = **202218.7448978875**
- p-value = **0.0**
- Since the p-value is far **smaller than 0.05**, we reject the null hypothesis H_0 that the returns are normally distributed.
- **Interpretation:** the return series exhibits **fat tails** (excess kurtosis) and **skewness**, deviating strongly from a normal distribution.

```
# Jarque-Bera Test
jb_test = jarque_bera(returns)
print("\nJarque-Bera Test:")
print("Statistic =", jb_test.statistic, ", p-value =", jb_test.pvalue)
```

```
Jarque-Bera Test:
Statistic = 202218.7448978875 , p-value = 0.0
```

Ex. 1 (d)

Annualized Return:

$$R_{annual} = R_{daily} \times 250$$

Annualized Volatility:

$$\sigma_{annual} = \sigma_{daily} \times \sqrt{250}$$

```
# Split into periods
before_2000 = df.loc[:'1999-12-31']['Daily Return'].dropna()
after_2000 = df.loc['2000-01-01':]['Daily Return'].dropna()
whole_period = df['Daily Return'].dropna()

# Calculate mean and std
results = pd.DataFrame({
    'Average Daily Return (%)': [
        whole_period.mean() * 100,
        before_2000.mean() * 100,
        after_2000.mean() * 100
    ],
    'Daily Std. Deviation (%)': [
        whole_period.std() * 100,
        before_2000.std() * 100,
        after_2000.std() * 100
    ]
}, index=['Whole Period', 'Before 2000', 'After 2000'])

results
```

	Average Daily Return (%)	Daily Std. Deviation (%)
Whole Period	0.040803	1.096984
Before 2000	0.051348	0.944628
After 2000	0.030785	1.224279

Next steps: Generate code with results New interactive sheet

Ex. 1 (e)

Comparison and Comments:

- The **whole period** shows about **10.2% annualized return** with volatility around **17.3%**.
- **Before 2000**: higher return (**12.8%**) and lower volatility (**14.9%**). As a consequence, a more favorable risk-return tradeoff.
- **After 2000**: lower return (**7.7%**) and higher volatility (**19.4%**). As a consequence, markets became riskier and less rewarding.
- This reflects major crises after 2000 (dot-com bubble, 2008 financial crisis, COVID-19 crash, Russian-Ukranian conflict, etc.).
- Overall, the S&P 500 provides positive long-term returns, but volatility is much larger than the mean return, consistent with the high-risk nature of equities.

```
# Helper function to compute annualized return & volatility
def annualize_stats(daily_returns):
    avg_daily = daily_returns.mean()
    std_daily = daily_returns.std()

    avg_annual = avg_daily * 250
    std_annual = std_daily * np.sqrt(250)

    return avg_daily*100, std_daily*100, avg_annual*100, std_annual*100
```

```
# Whole, before 2000, after 2000
whole = df['Daily Return'].dropna()
before_2000 = df.loc[:'1999-12-31']['Daily Return'].dropna()
after_2000 = df.loc['2000-01-01':]['Daily Return'].dropna()

# Build results table
results_annualized = pd.DataFrame(
    [annualize_stats(whole),
     annualize_stats(before_2000),
     annualize_stats(after_2000)],
    columns=[
        'Avg Daily Return (%)',
        'Daily Std Dev (%)',
        'Annualized Return (%)',
        'Annualized Std Dev (%)'
    ],
    index=['Whole Period','Before 2000','After 2000']
)

results_annualized
```

	Avg Daily Return (%)	Daily Std Dev (%)	Annualized Return (%)	Annualized Std Dev (%)
Whole Period	0.040803	1.096984	10.200775	17.344836
Before 2000	0.051348	0.944628	12.837006	14.935874
After 2000	0.030785	1.224279	7.696274	19.357557

Next steps: Generate code with `results_annualized` New interactive sheet

Ex. 1 (f)

Conclusion:

- At the **5% level**, the data behaves fairly close to normal.
- At the **1% level**, we observe **far more extreme events** than predicted by the normal distribution.
- This confirms the presence of **fat tails** in financial returns: large crashes and rallies occur more frequently than Gaussian theory suggests.

```
# Daily returns (drop missing values)
returns = df['Daily Return'].dropna()

# Sample mean & std
mu = returns.mean()
sigma = returns.std()

# Thresholds under normality
z_5 = 1.96   # ~5% two-tailed
z_1 = 2.58   # ~1% two-tailed

thr_5_upper = mu + z_5 * sigma
thr_5_lower = mu - z_5 * sigma

thr_1_upper = mu + z_1 * sigma
thr_1_lower = mu - z_1 * sigma

# Actual counts
n_total = len(returns)
outside_5 = np.sum((returns < thr_5_lower) | (returns > thr_5_upper))
outside_1 = np.sum((returns < thr_1_lower) | (returns > thr_1_upper))

# Percentages
perc_5 = outside_5 / n_total * 100
perc_1 = outside_1 / n_total * 100

print("5% tails:")
print("Expected ≈ 5%, Actual ≈", perc_5, "%, Count =", outside_5)

print("\n1% tails:")
print("Expected ≈ 1%, Actual ≈", perc_1, "%, Count =", outside_1)
```

```
5% tails:
Expected ≈ 5%, Actual ≈ 4.88694962316541 %, Count = 616

1% tails:
Expected ≈ 1%, Actual ≈ 2.2292740975803254 %, Count = 281
```

Ex. 2 (a)

> We are asked to compute the variance of: $P = 0.5X + 0.3Y + 0.2Z$

For any linear combination of random variables:
$\mathrm{Var}(aX + bY + cZ) = a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y) + c^2\,\mathrm{Var}(Z) + 2ab\,\mathrm{Cov}(X,Y) + 2ac\,\mathrm{Cov}(X,Z) + 2bc\,\mathrm{Cov}(Y,Z)$
$\mathrm{Var}(X) = \mathrm{Var}(Y) = \mathrm{Var}(Z) = 1$
$\mathrm{Cov}(X,Y) = 0.8$
$\mathrm{Cov}(X,Z) = 0.8$
$\mathrm{Cov}(Y,Z) = \rho$
$\mathrm{Var}(P) = (0.5)^2 + (0.3)^2 + (0.2)^2 + 2(0.5)(0.3)(0.8) + 2(0.5)(0.2)(0.8) + 2(0.3)(0.2)\rho$
$\mathrm{Var}(P) = 0.38 + 0.24 + 0.16 + 0.12\rho$
$\mathrm{Var}(P) = 0.78 + 0.12\rho$

```
# Define symbols
rho = sp.symbols('rho')
a, b, c = 0.5, 0.3, 0.2

# Variances
Var_X = Var_Y = Var_Z = 1
Cov_XY = 0.8
Cov_XZ = 0.8
Cov_YZ = rho

# Formula for Var(P)
Var_P = (a**2 * Var_X) + (b**2 * Var_Y) + (c**2 * Var_Z) \
        + 2*a*b*Cov_XY + 2*a*c*Cov_XZ + 2*b*c*Cov_YZ

# Simplify expression
Var_P_simplified = sp.simplify(Var_P)
Var_P_simplified
```

$0.12\rho + 0.78$

Ex. 2 (b)

For any weight vector $W = (w_x, w_y, w_z)$, the variance is defined as the expected squared deviation and can never be negative.

- If the correlation matrix Σ is **positive definite**, then for all nonzero W: $\mathrm{Var}(P) > 0$.
- If the correlation matrix Σ is only **positive semidefinite**, then for some special W the variance may equal 0 (degenerate case).

The variance of P is always non-negative. It is strictly positive for all nonzero weights if the correlation matrix is positive definite.

Ex. 2 (c)

Let the weight vector be

$$W = \begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix},$$

and the correlation matrix be

$$\Sigma = \begin{bmatrix} 1 & 0.8 & 0.8 \\ 0.8 & 1 & \rho \\ 0.8 & \rho & 1 \end{bmatrix}.$$

Then the variance of $P = w_x X + w_y Y + w_z Z$ is

$$\mathrm{Var}(P) = W^\top \Sigma W$$

The Expression Form of P:

$$\mathrm{Var}(P) = w_x^2 + w_y^2 + w_z^2 + 2(0.8)\, w_x w_y + 2(0.8)\, w_x w_z + 2\rho\, w_y w_z$$

Ex. 2 (d)

For any weights W, the variance must satisfy:

$$\mathrm{Var}(P) = W^\top \Sigma W \;\geq\; 0$$

This condition means that the correlation matrix Σ must be:

- **Symmetric**
- **Positive semidefinite (PSD)**. So, all eigenvalues ≥ 0 (so that $W^\top \Sigma W \geq 0$ for all W)
- With 1's on the diagonal (since each variable has variance 1)

The correlation matrix Σ must be **positive semidefinite (PSD)**.

Double-click (or enter) to edit

Ex. 2 (e)

The correlation matrix is:

$$\Sigma = \begin{bmatrix} 1 & 0.8 & 0.8 \\ 0.8 & 1 & \rho \\ 0.8 & \rho & 1 \end{bmatrix}$$

For

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}, \quad \det(A) = a(ei - fh) - b(di - fg) + c(dh - eg).$$

Here

$$\Sigma = \begin{bmatrix} 1 & 0.8 & 0.8 \\ 0.8 & 1 & \rho \\ 0.8 & \rho & 1 \end{bmatrix}.$$

Apply the formula

$$\det(\Sigma) = a(ei - fh) - b(di - fg) + c(dh - eg)$$

1. First term:

$$a(ei - fh) = 1 \cdot (1 \cdot 1 - \rho \cdot \rho) = 1 - \rho^2$$

2. Second term:

$$-b(di - fg) = -0.8(0.8 \cdot 1 - \rho \cdot 0.8) = -0.8(0.8 - 0.8\rho) = -0.64 + 0.64\rho$$

3. Third term:

$$+c(dh - eg) = 0.8(0.8 \cdot \rho - 1 \cdot 0.8) = 0.8(0.8\rho - 0.8) = 0.64\rho - 0.64$$

Combine and simplify

$$\begin{aligned} \det(\Sigma) &= (1 - \rho^2) + (-0.64 + 0.64\rho) + (0.64\rho - 0.64) \\ &= 1 - \rho^2 - 1.28 + 1.28\rho \\ &= -\rho^2 + 1.28\rho - 0.28 \\ \det(\Sigma) &= -\rho^2 + 1.28\rho - 0.28 \end{aligned}$$

```
# Define the correlation matrix Σ
Sigma = sp.Matrix([
    [1, 0.8, 0.8],
    [0.8, 1, rho],
    [0.8, rho, 1]
])

# Show the matrix
print("Correlation matrix Σ:")
sp.pprint(Sigma)

# General formula for 3x3 determinant
# det(A) = a(ei - fh) - b(di - fg) + c(dh - eg)
a, b, c, d, e, f, g, h, i = sp.symbols('a b c d e f g h i')
det_formula = a*(e*i - f*h) - b*(d*i - f*g) + c*(d*h - e*g)
print("\nGeneral 3x3 determinant formula:")
sp.pprint(det_formula)

# Substitute values from Σ into the formula
subs_dict = {
    a: 1, b: 0.8, c: 0.8,
    d: 0.8, e: 1, f: rho,
    g: 0.8, h: rho, i: 1
}
det_Sigma_manual = det_formula.subs(subs_dict)
```

```
print("\nDeterminant:")
sp.pprint(sp.expand(det_Sigma_manual))
```

```
Correlation matrix Σ:
⎡ 1   0.8  0.8⎤
⎢             ⎥
⎢0.8   1    ρ ⎥
⎢             ⎥
⎣0.8   ρ    1 ⎦

General 3x3 determinant formula:
a·(e·i - f·h) - b·(d·i - f·g) + c·(d·h - e·g)

Determinant:
   2
- ρ  + 1.28·ρ - 0.28
```

We found:

$$\det(\Sigma) = -\rho^2 + 1.28\rho - 0.28 \;\; \geq \;\; 0$$

Multiply by -1 (flip inequality)

$$\rho^2 - 1.28\rho + 0.28 \;\; \leq \;\; 0$$

Find the roots

$$\rho = \frac{1.28 \pm \sqrt{1.28^2 - 4(0.28)}}{2}$$

Simplify inside the square root

$$= \frac{1.28 \pm \sqrt{1.6384 - 1.12}}{2} = \frac{1.28 \pm \sqrt{0.5184}}{2}$$

Evaluate

$$= \frac{1.28 \pm 0.72}{2}$$

So the two roots are:

$$\rho = 0.28 \quad \text{and} \quad \rho = 1$$

Since $\rho^2 - 1.28\rho + 0.28 \leq 0$, the solution lies **between the roots**:

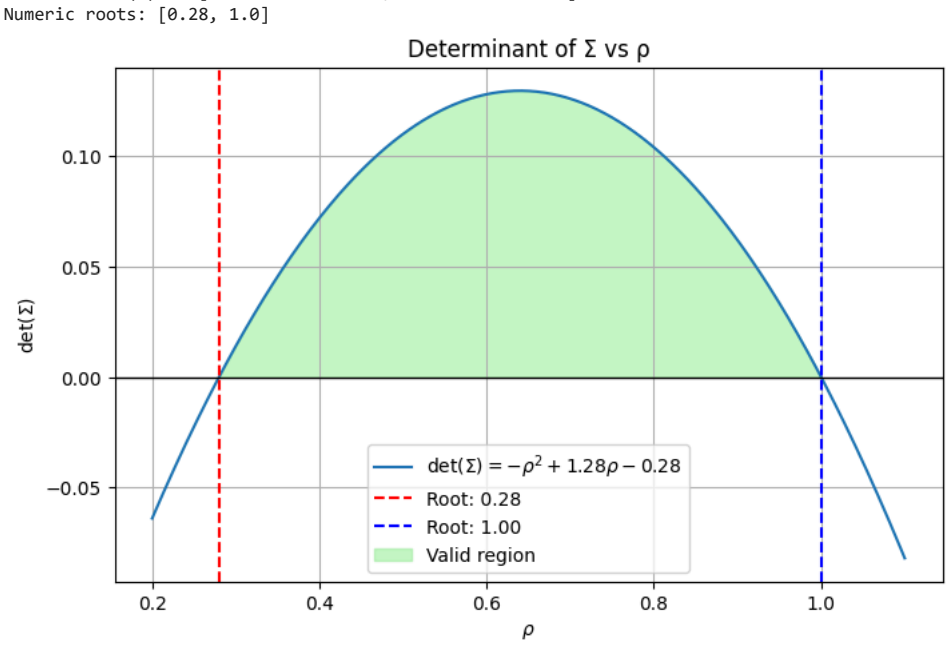
$$0.28 \leq \rho \leq 1$$

The valid range of ρ is

$$[0.28,\ 1]$$

```
# Determinant expression
det_Sigma = -rho**2 + 1.28*rho - 0.28
print("Determinant of Σ:", det_Sigma)

# Solve inequality det(Σ) >= 0
solution = sp.solve_univariate_inequality(det_Sigma >= 0, rho)
print("Valid range for rho:", solution)

# Solve quadratic equation
roots = sp.solve(sp.Eq(det_Sigma, 0), rho)
print("Roots of det(Σ)=0:", roots)

# Numeric evaluation of roots
roots_eval = [float(r) for r in roots]
print("Numeric roots:", roots_eval)

# Plot det(Σ) over [0.28, 1]
rho_vals = np.linspace(0.2, 1.1, 400)
det_vals = [-r**2 + 1.28*r - 0.28 for r in rho_vals]

plt.figure(figsize=(8,5))
plt.plot(rho_vals, det_vals, label=r"$\det(\Sigma) = -\rho^2 + 1.28\rho - 0.28$")
plt.axhline(0, color="black", linewidth=1)
plt.axvline(roots_eval[0], color="red", linestyle="--", label=f"Root: {roots_eval[0]:.2f}")
plt.axvline(roots_eval[1], color="blue", linestyle="--", label=f"Root: {roots_eval[1]:.2f}")
plt.fill_between(rho_vals, det_vals, 0, where=(np.array(det_vals)>=0), color="lightgreen", alpha=0.5, label="Valid region")

plt.title("Determinant of Σ vs ρ")
plt.xlabel(r"$\rho$")
plt.ylabel(r"$\det(\Sigma)$")
plt.legend()
plt.grid(True)
plt.show()
```

```
Determinant of Σ: -rho**2 + 1.28*rho - 0.28
Valid range for rho: (0.28 <= rho) & (rho <= 1.0)
Roots of det(Σ)=0: [0.280000000000000, 1.00000000000000]
Numeric roots: [0.28, 1.0]
```



Ex. 3 (a)

We are given the covariance matrix:

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}.$$

Eigenvalues λ satisfy:

$$\det(\Sigma - \lambda I) = 0,$$
$$\det \begin{bmatrix} \sigma_1^2 - \lambda & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 - \lambda \end{bmatrix} = 0.$$

For a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, we have $\det = ad - bc$. So:

$$\det(\Sigma - \lambda I) = (\sigma_1^2 - \lambda)(\sigma_2^2 - \lambda) - (\rho\sigma_1\sigma_2)^2.$$

Expand $(\sigma_1^2 - \lambda)(\sigma_2^2 - \lambda)$:

$$(\sigma_1^2 - \lambda)(\sigma_2^2 - \lambda) = \sigma_1^2\sigma_2^2 - \sigma_1^2\lambda - \sigma_2^2\lambda + \lambda^2.$$

So:

$$\det(\Sigma - \lambda I) = \lambda^2 - (\sigma_1^2 + \sigma_2^2)\lambda + \sigma_1^2\sigma_2^2 - \rho^2\sigma_1^2\sigma_2^2.$$

Simplify constant term:

$$\det(\Sigma - \lambda I) = \lambda^2 - (\sigma_1^2 + \sigma_2^2)\lambda + \sigma_1^2\sigma_2^2(1 - \rho^2).$$

For quadratic $a\lambda^2 + b\lambda + c = 0$ with

- $a = 1$
- $b = -(\sigma_1^2 + \sigma_2^2)$
- $c = \sigma_1^2\sigma_2^2(1 - \rho^2)$

The discriminant is:

$$\Delta = b^2 - 4ac = (\sigma_1^2 + \sigma_2^2)^2 - 4\sigma_1^2\sigma_2^2(1 - \rho^2).$$

Simplify:

$$\Delta = (\sigma_1^2 - \sigma_2^2)^2 + 4\rho^2\sigma_1^2\sigma_2^2 \ \geq 0.$$

Therefore:

$$\lambda_{1,2} = \frac{\sigma_1^2 + \sigma_2^2}{2} \ \pm \ \frac{1}{2}\sqrt{(\sigma_1^2 - \sigma_2^2)^2 + 4\rho^2\sigma_1^2\sigma_2^2},$$

with $\lambda_1 \geq \lambda_2 \geq 0$.

```
# Symbols
s1, s2, r, lam = sp.symbols('s1 s2 r lam', real=True)

# Covariance matrix Σ
Sigma = sp.Matrix([[s1**2, r*s1*s2],
                   [r*s1*s2, s2**2]])
print("Covariance matrix Σ =")
display(Sigma)

# Eigenvalues (Sympy computes automatically)
eigvals = Sigma.eigenvals()   # dictionary {eigenvalue: multiplicity}
eigvals_list = list(eigvals.keys())

print("\nEigenvalues (symbolic):")
for ev in eigvals_list:
    display(ev)
```

```
Covariance matrix Σ =
```

$$\begin{bmatrix} s_1^2 & rs_1s_2 \\ rs_1s_2 & s_2^2 \end{bmatrix}$$

```
Eigenvalues (symbolic):
```

$$\frac{s_1^2}{2} + \frac{s_2^2}{2} - \frac{\sqrt{4r^2s_1^2s_2^2 + s_1^4 - 2s_1^2s_2^2 + s_2^4}}{2}$$

$$\frac{s_1^2}{2} + \frac{s_2^2}{2} + \frac{\sqrt{4r^2s_1^2s_2^2 + s_1^4 - 2s_1^2s_2^2 + s_2^4}}{2}$$

Ex. 3 (b)

We now compute the eigenvectors γ_1, γ_2:

Assume $\gamma = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ such that:

$$\Sigma\gamma = \lambda\gamma.$$

This gives the system:

$$(\sigma_1^2 - \lambda)v_1 + \rho\sigma_1\sigma_2 v_2 = 0,$$
$$\rho\sigma_1\sigma_2 v_1 + (\sigma_2^2 - \lambda)v_2 = 0.$$

Because we have $\gamma_i'\gamma_i = 1$, we can express the eigenvector as

$\gamma = \begin{pmatrix} \cos\phi \\ \sin\phi \end{pmatrix}$.

Then we have the two conditions from:

$\Sigma\gamma = \lambda\gamma$:

This way, $||\gamma|| = 1$ is automatic. We require

$$\Sigma\gamma = \lambda\gamma$$

Substitute γ:

$$\begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix} \begin{pmatrix} \cos\phi \\ \sin\phi \end{pmatrix} = \lambda \begin{pmatrix} \cos\phi \\ \sin\phi \end{pmatrix}$$

This gives two scalar equations:

$$\sigma_1^2\cos\phi + \rho\sigma_1\sigma_2\sin\phi = \lambda\cos\phi \tag{1}$$
$$\rho\sigma_1\sigma_2\cos\phi + \sigma_2^2\sin\phi = \lambda\sin\phi \tag{2}$$

Dividing through and equating leads to:

$$\frac{\sigma_1^2\cos\phi + \rho\sigma_1\sigma_2\sin\phi}{\cos\phi} \ = \ \frac{\rho\sigma_1\sigma_2\cos\phi + \sigma_2^2\sin\phi}{\sin\phi}.$$

Multiply through by $\sin\phi\cos\phi$:

$$\sigma_1^2\sin\phi\cos\phi + \rho\sigma_1\sigma_2\sin^2\phi = \sigma_2^2\sin\phi\cos\phi + \rho\sigma_1\sigma_2\cos^2\phi$$

Move the $\sin\phi\cos\phi$ terms to the left, and the squared terms to the right:

$$\sigma_1^2\sin\phi\cos\phi - \sigma_2^2\sin\phi\cos\phi = \rho\sigma_1\sigma_2\cos^2\phi - \rho\sigma_1\sigma_2\sin^2\phi$$

Bring terms together:

$$(\sigma_1^2 - \sigma_2^2)\sin\phi\cos\phi = \rho\sigma_1\sigma_2(\cos^2\phi - \sin^2\phi)$$

Use trig identities

- $\sin\phi\cos\phi = \frac{1}{2}\sin(2\phi)$
- $\cos^2\phi - \sin^2\phi = \cos(2\phi)$

So:

$$(\sigma_1^2 - \sigma_2^2)\tfrac{1}{2}\sin(2\phi) = \rho\sigma_1\sigma_2\cos(2\phi)$$

Divide both sides by $\cos(2\phi)$

$$\frac{(\sigma_1^2 - \sigma_2^2)}{2} \cdot \frac{\sin(2\phi)}{\cos(2\phi)} = \rho\sigma_1\sigma_2$$

Recognize tangent

Since $\frac{\sin(2\phi)}{\cos(2\phi)} = \tan(2\phi)$:

$$\frac{(\sigma_1^2 - \sigma_2^2)}{2}\tan(2\phi) = \rho\sigma_1\sigma_2$$

Thus, we obtain:

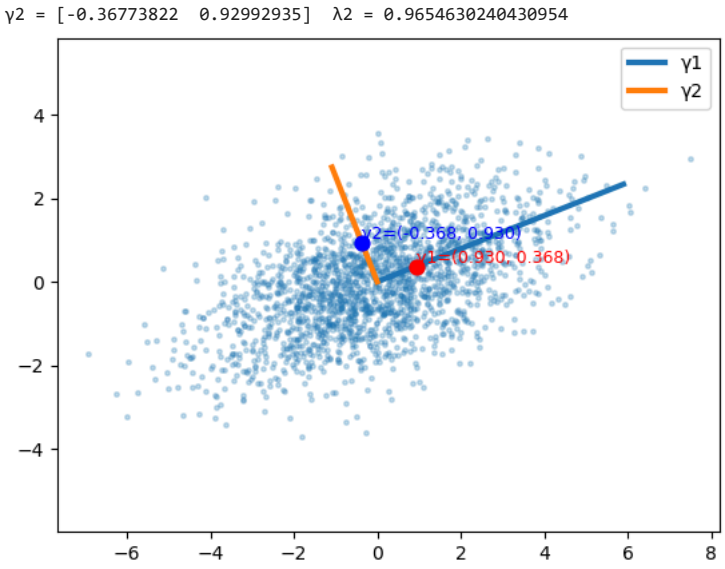
$$\tan(2\phi) = \frac{2\rho\sigma_1\sigma_2}{\sigma_1^2 - \sigma_2^2}.$$

The two orthonormal eigenvectors are:

$$\gamma_1 = \begin{pmatrix} \cos\phi \\ \sin\phi \end{pmatrix}, \qquad \gamma_2 = \begin{pmatrix} -\sin\phi \\ \cos\phi \end{pmatrix}.$$

```
# Parameters
sigma1, sigma2, rho = 2.0, 1.2, 0.5
Sigma = np.array([[sigma1**2, rho*sigma1*sigma2],
                  [rho*sigma1*sigma2, sigma2**2]])

# Angle ϕ from closed form
phi = 0.5 * np.arctan2(2*rho*sigma1*sigma2, sigma1**2 - sigma2**2)

# Eigenvectors
gamma1 = np.array([np.cos(phi), np.sin(phi)])
gamma2 = np.array([-np.sin(phi), np.cos(phi)])

# Eigenvalues
lambda1 = gamma1 @ Sigma @ gamma1
lambda2 = gamma2 @ Sigma @ gamma2

print("ϕ =", phi)
print("γ1 =", gamma1, " λ1 =", lambda1)
print("γ2 =", gamma2, " λ2 =", lambda2)

# Plot samples and principal axes
X = np.random.multivariate_normal([0,0], Sigma, size=2000)
plt.scatter(X[:,0], X[:,1], s=6, alpha=0.25)
plt.plot([0, gamma1[0]*3*np.sqrt(lambda1)], [0, gamma1[1]*3*np.sqrt(lambda1)], lw=3, label="γ1")
plt.plot([0, gamma2[0]*3*np.sqrt(lambda2)], [0, gamma2[1]*3*np.sqrt(lambda2)], lw=3, label="γ2")

# Add points for γ1 and γ2 (unit eigenvectors)
plt.scatter(gamma1[0], gamma1[1], color="red", s=60, zorder=5)
plt.scatter(gamma2[0], gamma2[1], color="blue", s=60, zorder=5)

# Annotate with values
plt.text(gamma1[0], gamma1[1], f"γ1=({gamma1[0]:.3f}, {gamma1[1]:.3f})",
         fontsize=9, color="red", ha="left", va="bottom")
plt.text(gamma2[0], gamma2[1], f"γ2=({gamma2[0]:.3f}, {gamma2[1]:.3f})",
         fontsize=9, color="blue", ha="left", va="bottom")

plt.axis("equal"); plt.legend(); plt.show()
```

```
ϕ = 0.3765756404810972
γ1 = [0.92992935 0.36773822]  λ1 = 4.474536975956905
γ2 = [-0.36773822  0.92992935]  λ2 = 0.9654630240430954
```



Ex. 3 (c)

Demonstrate that:

$$\Sigma = \sum_{i=1}^{m} \lambda_i \gamma_i \gamma_i'$$

Because of $\Sigma\gamma_i = \lambda_i\gamma_i$, we have

$$\Sigma\gamma_i\gamma_i' = \lambda_i\gamma_i\gamma_i'$$

then

$$\sum_{i=1}^{m} \Sigma\gamma_i\gamma_i' = \sum_{i=1}^{m} \lambda_i\gamma_i\gamma_i'.$$

For

$$\sum_{i=1}^{m} \gamma_i\gamma_i' = \gamma_1\gamma_1' + \gamma_2\gamma_2' + \cdots + \gamma_m\gamma_m',$$

let $G = [\gamma_1, \gamma_2, \cdots, \gamma_m]$, G is an orthogonal matrix.
Then

$$GG' = G^{-1}G = I,$$

so

$$\sum_{i=1}^{m} \gamma_i\gamma_i' = \gamma_1\gamma_1' + \gamma_2\gamma_2' + \cdots + \gamma_m\gamma_m' = I.$$

Therefore,

$$\Sigma = \sum_{i=1}^{m} \lambda_i\gamma_i\gamma_i'.$$

Ex. 4 (a)

The probability density function of an exponential random variable X with parameter $\lambda > 0$ is

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0 \quad \Rightarrow \quad f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

The moment generating function (MGF) is

$$M_X(t) = \mathbb{E}[e^{tX}] = \int_0^\infty e^{tx} f_X(x)\, dx = \int_0^\infty e^{tx}\, \lambda e^{-\lambda x}\, dx = \lambda \int_0^\infty e^{-(\lambda - t)x}\, dx.$$

This integral converges only if $t < \lambda$.
Find an antiderivative

$$\int e^{-(\lambda-t)x}\, dx = -\frac{1}{\lambda - t}\, e^{-(\lambda-t)x} + C.$$

Apply bounds

$$\int_0^b e^{-(\lambda-t)x}\, dx = \left[-\frac{1}{\lambda-t} e^{-(\lambda-t)x}\right]_{x=0}^{x=b} = -\frac{1}{\lambda-t} e^{-(\lambda-t)b} + \frac{1}{\lambda-t}.$$

Take the limit as $b \Rightarrow \infty$ (using $t < \lambda$ so $e^{-(\lambda-t)b} \Rightarrow 0$)

$$\int_0^\infty e^{-(\lambda-t)x}\, dx = \lim_{b\to\infty} \left(\frac{1}{\lambda-t} - \frac{1}{\lambda-t} e^{-(\lambda-t)b} \right) = \frac{1}{\lambda-t}.$$

The moment generating function of $X \sim \text{Exponential}(\lambda)$ is

$$M_X(t) = \frac{\lambda}{\lambda - t}, \quad t < \lambda.$$

Ex. 4 (b)

Checking Defenition of MGF at $t = 0$

$$M_X(0) = \frac{\lambda}{\lambda - 0} = \frac{\lambda}{\lambda} = 1$$

First derivative (with respect to t)

$$M_X'(t) = \frac{d}{dt}\left(\frac{\lambda}{\lambda - t}\right) = \frac{\lambda}{(\lambda - t)^2}.$$

At $t = 0$:

$$M_X'(0) = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}.$$

So the mean is:

$$\mathbb{E}[X] = M_X'(0) = \frac{1}{\lambda}.$$

Second derivative (with respect to t)

$$M_X''(t) = \frac{d}{dt}\left(\frac{\lambda}{(\lambda - t)^2}\right) = \frac{2\lambda}{(\lambda - t)^3}.$$

At $t = 0$:

$$M_X''(0) = \frac{2\lambda}{\lambda^3} = \frac{2}{\lambda^2}.$$

So the variance is:

$$\text{Var}(X) = M_X''(0) - \left(M_X'(0)\right)^2 = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}.$$

Ex. 4 (c)

We want the probability that X exceeds a fixed real $t > 0$:

$$P(X > t) = \int_t^\infty f_X(x)\, dx.$$

Since the PDF is

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0,$$

we have

$$P(X > t) = \int_t^\infty \lambda e^{-\lambda x}\, dx.$$

Find antiderivative

$$\int \lambda e^{-\lambda x}\, dx = -e^{-\lambda x} + C.$$

Apply bounds

$$\int_t^\infty \lambda e^{-\lambda x}\, dx = \left[-e^{-\lambda x}\right]_{x=t}^{x=\infty} = 0 - \left(-e^{-\lambda t}\right).$$

Simplify

$$P(X > t) = e^{-\lambda t}.$$

For $X \sim \text{Exponential}(\lambda)$ and any $t > 0$:

$$P(X > t) = e^{-\lambda t}.$$

Ex. 4 (d)

We want to show:

$$P(X > s + t \mid X > s) = P(X > t), \quad \forall\, s, t > 0.$$

Definition of conditional probability

$$P(X > s + t \mid X > s) = \frac{P(X > s + t \,\cap\, X > s)}{P(X > s)}.$$

But the event $\{X > s + t\}$ automatically implies $\{X > s\}$.
So:

$$P(X > s + t \mid X > s) = \frac{P(X > s + t)}{P(X > s)}.$$

Use the survival function From part (c) we know:

$$P(X > u) = e^{-\lambda u}, \quad u > 0.$$

So:

$$P(X > s + t \mid X > s) = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}}.$$

Simplify

$$P(X > s + t \mid X > s) = e^{-\lambda t}.$$

But this is exactly:

$$P(X > t).$$

The exponential distribution satisfies the **memoryless property**:

$$P(X > s + t \mid X > s) = P(X > t), \quad \forall\, s, t > 0.$$

Ex. 4 (e)

Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables with distribution

$$X_i \sim \text{Exponential}(\lambda).$$

We want to show that

$$Y = \min\{X_1, X_2, \ldots, X_n\}$$

is also exponentially distributed.

Tail probability of the minimum. By definition:

$$P(Y > t) = P(\min\{X_1, \ldots, X_n\} > t).$$

But the minimum exceeds t if **all variables exceed** t.

So:

$$P(Y > t) = P(X_1 > t, X_2 > t, \ldots, X_n > t).$$

Use independence Since the X_i are independent:

$$P(Y > t) = \prod_{i=1}^{n} P(X_i > t).$$

Recall exponential tail probability From part (c):

$$P(X_i > t) = e^{-\lambda t}.$$

Thus:

$$P(Y > t) = \left(e^{-\lambda t}\right)^n = e^{-n\lambda t}.$$

Identify the distribution The survival function of Y is:

$$P(Y > t) = e^{-n\lambda t}, \quad t \geq 0.$$

This is exactly the survival function of an exponential random variable with parameter $n\lambda$.

$$Y = \min\{X_1, \ldots, X_n\} \ \sim \ \text{Exponential}(n\lambda).$$

Ex. 5 (a)

Let $X \sim \text{Lognormal}(\mu, \sigma^2)$. Its PDF is:

$$f(x; \mu, \sigma) = \begin{cases} \dfrac{1}{\sqrt{2\pi}\sigma x} \exp\left(-\dfrac{(\ln x - \mu)^2}{2\sigma^2}\right), & x > 0, \\ 0, & x \leq 0. \end{cases}$$

Mean

$$E[X] = \int_0^\infty x f(x)\, dx$$

Substitute $y = \ln x \implies x = e^y$, $dx = e^y dy$:

$$E[X] = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} \exp\left(y - \frac{(y-\mu)^2}{2\sigma^2}\right) dy$$

Complete the square in the exponent:

$$y - \frac{(y-\mu)^2}{2\sigma^2} = -\frac{(y - (\mu + \sigma^2))^2}{2\sigma^2} + \mu + \tfrac{1}{2}\sigma^2$$

Thus,

$$E[X] = e^{\mu + \frac{1}{2}\sigma^2}$$

Second Moment

$$E[X^2] = \int_0^\infty x^2 f(x)\, dx = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{2y} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right) dy$$

By the same method,

$$E[X^2] = e^{2\mu + 2\sigma^2}$$

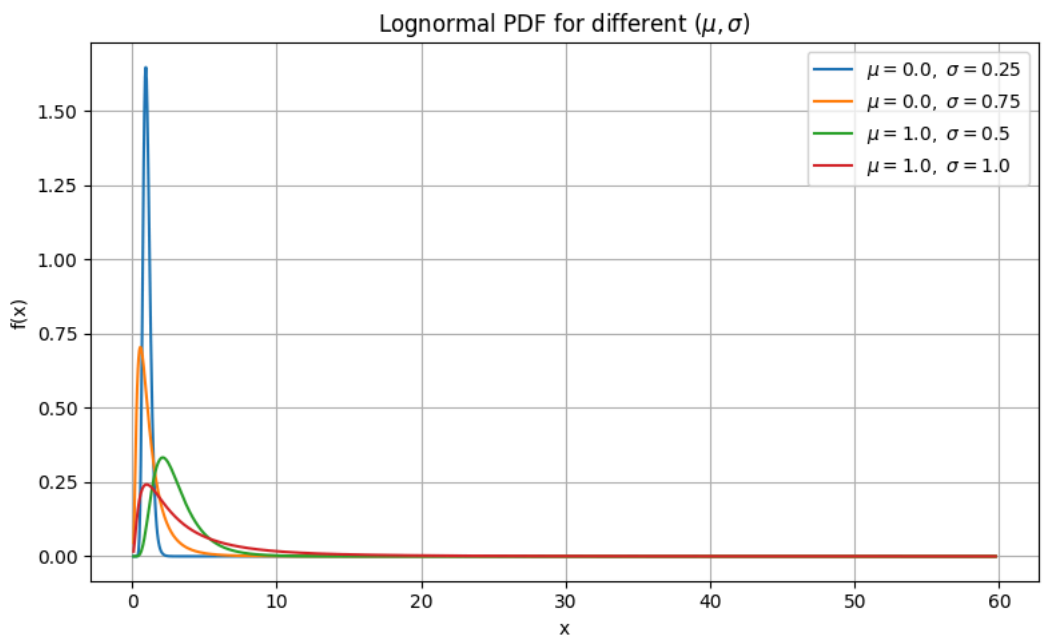
Variance

$$\text{Var}(X) = E[X^2] - (E[X])^2 = e^{2\mu+2\sigma^2} - e^{2\mu+\sigma^2}$$
$$\text{Var}(X) = \left(e^{\sigma^2} - 1\right) e^{2\mu+\sigma^2}$$

Ex. 5 (b)

```
# Choose parameter pairs (μ, σ)
params = [(0.0, 0.25), (0.0, 0.75), (1.0, 0.5), (1.0, 1.0)]

# Build a common x-range using extreme quantiles across all params
q_lo, q_hi = 1e-3, 1 - 1e-3
lo = min(lognorm.ppf(q_lo, s=s, scale=np.exp(m)) for m, s in params)
hi = max(lognorm.ppf(q_hi, s=s, scale=np.exp(m)) for m, s in params)
x = np.linspace(lo, hi, 1200)

# Plot PDFs
plt.figure(figsize=(8,5))
for m, s in params:
    pdf = lognorm.pdf(x, s=s, scale=np.exp(m))
    plt.plot(x, pdf, label=fr"$\mu={m},\ \sigma={s}$")

plt.title("Lognormal PDF for different $(\mu,\sigma)$")
plt.xlabel("x"); plt.ylabel("f(x)")
plt.grid(True); plt.legend()
plt.tight_layout(); plt.show()
```

```
<>:16: SyntaxWarning: invalid escape sequence '\m'
<>:16: SyntaxWarning: invalid escape sequence '\m'
/tmp/ipython-input-2032304428.py:16: SyntaxWarning: invalid escape sequence '\m'
  plt.title("Lognormal PDF for different $(\mu,\sigma)$")
```



Ex. 6

Given that the radius of the circle is 1. Suppose the diameter lies on the X-axis from $A(-1, 0)$ to $B(1, 0)$.

The semicircle arc is above the X-axis with equation:

$$y = \sqrt{1-x^2}, \quad x \in [-1, 1].$$

A random point P is to be chosen on the **circumference** to form a right triangle $\triangle APB$.

The circumference of the boundary is:

$$\frac{2\pi r}{2} + 2r = \pi + 2.$$

We set the coordinates of P (randomly) as $(\cos\theta, \sin\theta)$.

- For $\theta \in [0, \pi]$: P is on the semicircular arc ($y \geq 0$).
- For $\theta \in [\pi, 2\pi]$: P is on the diameter ($y = 0$).

For $\theta \in [0, \pi]$:

$$PA = \sqrt{(\cos\theta - (-1))^2 + (\sin\theta - 0)^2} = \sqrt{(\cos\theta + 1)^2 + \sin^2\theta} = \sqrt{2 + 2\cos\theta} = 2|\cos(\theta/2)| = 2\cos(\theta/2)$$

$$(\theta \in [0, \pi] \to \frac{\theta}{2} \in [0, \pi/2] \to \cos(\theta/2) \geq 0)$$

$$PB = \sqrt{(\cos\theta - 1)^2 + (\sin\theta - 0)^2} = \sqrt{(\cos\theta - 1)^2 + \sin^2\theta} = \sqrt{2 - 2\cos\theta} = 2|\sin(\theta/2)| = 2\sin(\theta/2)$$

$$(\theta \in [0, \pi] \to \frac{\theta}{2} \in [0, \pi/2] \to \sin(\theta/2) \geq 0)$$

The area of the triangle when $\theta \in [0, \pi]$:

$$Area = \tfrac{1}{2} \times PA \times PB = \tfrac{1}{2} \times 4\sin(\tfrac{\theta}{2})\cos(\tfrac{\theta}{2}) = 2\sin(\tfrac{\theta}{2})\cos(\tfrac{\theta}{2}) = \sin\theta$$

When $\theta \in [\pi, 2\pi]$, P lies on the diameter, the area should be 0.

Now we can calculate for the expected area:

$$E[Area] = E[Area \mid \theta \in [0, \pi]] \cdot P(\theta \in [0, \pi]) + E[Area \mid \theta \in [\pi, 2\pi]] \cdot P(\theta \in [\pi, 2\pi])$$

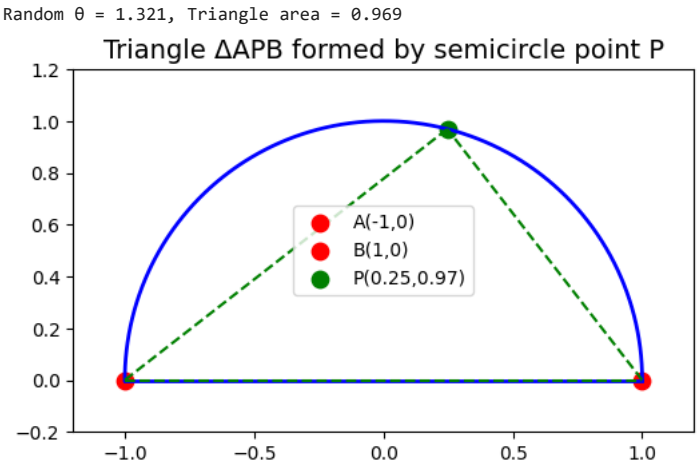
$$E[Area \mid \theta \in [0, \pi]] = \frac{1}{\pi}\int_0^\pi \sin\theta \, d\theta = \frac{1}{\pi}[-\cos\theta]_0^\pi = \frac{1}{\pi}[-\cos\pi - (-\cos 0)] = \frac{1}{\pi}[1 + 1] = \frac{2}{\pi}$$

$$E[Area \mid \theta \in [\pi, 2\pi]] = 0$$

$$E[Area] = \left(\frac{2}{\pi}\right) \cdot \tfrac{1}{2} + 0 \cdot \tfrac{1}{2} = \frac{2}{2\pi} = \frac{1}{\pi}$$

```
# Circle parameters
r = 1
theta = np.linspace(0, np.pi, 300)

# Semicircle coordinates
x = np.cos(theta)
y = np.sin(theta)

# Endpoints of diameter
A = (-1, 0)
B = (1, 0)

# Pick a random point P on the semicircle
theta_rand = np.random.uniform(0, np.pi)
P = (np.cos(theta_rand), np.sin(theta_rand))

# Compute area of triangle APB
area = 0.5 * abs((A[0]*(B[1]-P[1]) + B[0]*(P[1]-A[1]) + P[0]*(A[1]-B[1])))
print(f"Random θ = {theta_rand:.3f}, Triangle area = {area:.3f}")

# Plot
fig, ax = plt.subplots(figsize=(6,6))
ax.plot(x, y, 'b', linewidth=2)  # semicircle
ax.plot([-1,1], [0,0], 'b', linewidth=2)  # diameter
ax.scatter(*A, color="red", s=80, label="A(-1,0)")
ax.scatter(*B, color="red", s=80, label="B(1,0)")
ax.scatter(*P, color="green", s=80, label=f"P({P[0]:.2f},{P[1]:.2f})")

# Draw triangle APB
ax.plot([A[0], P[0]], [A[1], P[1]], 'g--')
ax.plot([B[0], P[0]], [B[1], P[1]], 'g--')
ax.plot([A[0], B[0]], [A[1], B[1]], 'g--')

# Style
ax.set_aspect('equal')
ax.set_xlim(-1.2, 1.2)
ax.set_ylim(-0.2, 1.2)
ax.set_title("Triangle ΔAPB formed by semicircle point P", fontsize=14)
ax.legend()
plt.show()
```

```
Random θ = 1.321, Triangle area = 0.969
```



Ex. 7

Double-click (or enter) to edit

The **Touchard formula** gives the discordant permutation numbers A_n:

$A_n = \sum_{k=0}^{n} (-1)^k \frac{2n}{2n-k} \binom{2n-k}{k} (n-k)!$.

Compute A_3

$$
\begin{aligned}
A_3 &= \sum_{k=0}^{3} (-1)^k \frac{6}{6-k} \binom{6-k}{k} (3-k)! \\
&= (+)\frac{6}{6}\binom{6}{0}(3)! \;-\; \frac{6}{5}\binom{5}{1}(2)! \;+\; \frac{6}{4}\binom{4}{2}(1)! \;-\; \frac{6}{3}\binom{3}{3}(0)! \\
&= 6 - 12 + 9 - 2 \\
&= 1.
\end{aligned}
$$

So we find that $A_3 = 1$.

This is a version of the **ménage problem**.

- With men fixed in order, women must be permuted into n slots so that wife W_i does not sit in the two slots adjacent to M_i.
- This is a permutation problem with forbidden positions, solved by rook polynomials.

The number of valid women placements is

$$D_n = \sum_{k=0}^{n} (-1)^k \, r_k \, (n-k)!,$$

where

$$r_k = \frac{2n}{2n-k}\binom{2n-k}{k},$$

is the number of ways to place k non-attacking rooks on the forbidden board.

Including men's permutations (with one man fixed to remove rotation):

$$N(n) = (n-1)!\, D_n.$$

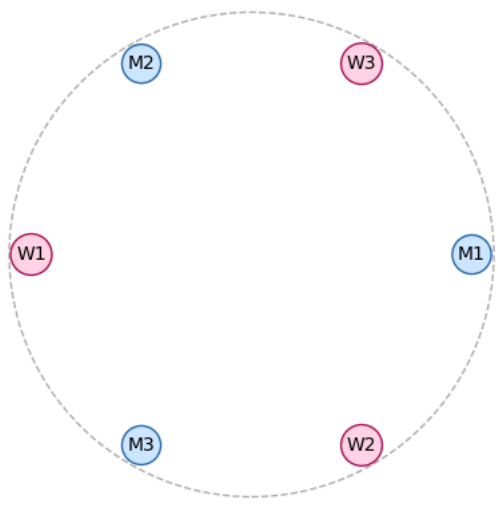
$$\boxed{N(n) = (n-1)! \sum_{k=0}^{n} (-1)^k \frac{2n}{2n-k}\binom{2n-k}{k}(n-k)!}$$

```
import itertools, numpy as np
import matplotlib.pyplot as plt, matplotlib.animation as animation
from IPython.display import HTML
import ipywidgets as widgets

def visualize_couples(n=3):
    men   = [f"M{i+1}" for i in range(n)]
    women = [f"W{i+1}" for i in range(n)]
    men_seats   = list(range(0, 2*n, 2))
    women_seats = list(range(1, 2*n, 2))

    angles = np.linspace(0, 2*np.pi, 2*n, endpoint=False)
    pos = np.c_[np.cos(angles), np.sin(angles)]

    def is_valid_women_perm(perm):
        for seat_index, wife in zip(women_seats, perm):
            wife_id = int(wife[1:])  # W3 -> 3
            husband = men_seats[wife_id-1]
            if husband in ((seat_index-1)%(2*n), (seat_index+1)%(2*n)):
                return False
        return True

    valid_arrangements = [perm for perm in itertools.permutations(women)
                          if is_valid_women_perm(perm)]
    print(f"Number of valid seatings for n={n}: {len(valid_arrangements)}")

    if not valid_arrangements:
        print("No valid arrangements."); return

    fig, ax = plt.subplots(figsize=(6,6))

    def draw_arrangement(perm):
        ax.clear(); ax.set_aspect('equal'); ax.axis('off')
        ax.set_xlim(-1.4,1.4); ax.set_ylim(-1.4,1.4)
        ax.add_artist(plt.Circle((0,0),1.1,fill=False,ls='--',alpha=0.3))

        for i,m in enumerate(men):
            x,y = pos[men_seats[i]]
            ax.text(x,y,m,ha='center',va='center',
                    bbox=dict(boxstyle="circle,pad=0.3",facecolor="#cfe8ff",edgecolor="#2b6cb0"))
        for i,w in enumerate(perm):
            x,y = pos[women_seats[i]]
            ax.text(x,y,w,ha='center',va='center',
                    bbox=dict(boxstyle="circle,pad=0.3",facecolor="#ffd6e7",edgecolor="#ad1457"))
        ax.set_title("Valid seating (alternating sexes, distinct seatings up to rotation)")

    def update(frame):
        draw_arrangement(valid_arrangements[frame % len(valid_arrangements)])
        return []

    anim = animation.FuncAnimation(fig, update, frames=len(valid_arrangements),
                                   interval=1500, blit=False, repeat=True)
    display(HTML(anim.to_jshtml()))
    plt.close(fig)

widgets.interact(visualize_couples,
                 n=widgets.IntSlider(value=3, min=3, max=5, step=1, description="n couples"))
```

n couples 3

```
Number of valid seatings for n=3: 1
```

Valid seating (alternating sexes, distinct seatings up to rotation)



Once Loop Reflect

```
visualize_couples
def visualize_couples(n=3)
```

```
<no docstring>
```

Ex. 8 (a)

Suppose we have the correlation matrix

$$A = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}, \qquad -1 < \rho < 1.$$

The Cholesky decomposition seeks a triangular matrix L with positive diagonal entries such that $A = LL^\top$.

Lower–triangular form

$$L = \begin{bmatrix} 1 & 0 \\ \rho & \sqrt{1-\rho^2} \end{bmatrix}.$$

Check:

$$LL^\top = \begin{bmatrix} 1 & 0 \\ \rho & \sqrt{1-\rho^2} \end{bmatrix} \begin{bmatrix} 1 & \rho \\ 0 & \sqrt{1-\rho^2} \end{bmatrix} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}.$$

Upper–triangular form

Equivalently, one can write

$$U = L^\top = \begin{bmatrix} 1 & \rho \\ 0 & \sqrt{1-\rho^2} \end{bmatrix},$$

so that

$$A = U^\top U.$$

Ex. 8 (b)

Given a real, symmetric, positive–definite matrix

$$A \in \mathbb{R}^{n\times n},$$

the Cholesky decomposition finds a **lower triangular** matrix L with positive diagonal entries such that

$$A = LL^\top$$

Algorithm (step–by–step) For $i = 1, 2, \ldots, n$ (rows):

- Compute the diagonal entry $L_{ii} = \sqrt{A_{ii} - \sum_{k=1}^{i-1} L_{ik}^2}$ For each $j = i+1, i+2, \ldots, n$ (rows below the diagonal):
- Compute the off–diagonal entries $L_{ji} = \frac{1}{L_{ii}}\left(A_{ji} - \sum_{k=1}^{i-1} L_{jk}L_{ik}\right)$ All entries above the diagonal are zero:

  $$L_{ij} = 0 \quad \text{for } j > i.$$

  Properties
- The complexity is $O(n^3)$ (similar to Gaussian elimination).
- L is unique if we require $L_{ii} > 0$.
- Numerical stability requires A to be strictly positive–definite.

Ex. 8 (c)

```
from scipy.linalg import cholesky as scipy_chol

def cholesky_lower(A, tol=1e-12):
    A = np.array(A, float); n = A.shape[0]
    L = np.zeros_like(A)
    for i in range(n):
        s = np.dot(L[i,:i], L[i,:i])
        v = A[i,i]-s
        if v<=tol: raise np.linalg.LinAlgError("Not positive definite")
        L[i,i] = np.sqrt(v)
        for j in range(i+1,n):
            s = np.dot(L[j,:i], L[i,:i])
            L[j,i] = (A[j,i]-s)/L[i,i]
    return L

# Build SPD test matrices
rho=0.6; A2=np.array([[1,rho],[rho,1]])
B3=np.random.randn(3,3); A3=B3@B3.T+0.5*np.eye(3)
B5=np.random.randn(5,5); A5=B5@B5.T+1.0*np.eye(5)
tests=[("2x2",A2),("3x3",A3),("5x5",A5)]

for name,A in tests:
    Lc=cholesky_lower(A)
    Ln=np.linalg.cholesky(A)
```

```
    Ls=scipy_chol(A,lower=True)
    err_c=np.linalg.norm(A-Lc@Lc.T,'fro')
    err_n=np.linalg.norm(A-Ln@Ln.T,'fro')
    err_s=np.linalg.norm(A-Ls@Ls.T,'fro')

    print(f"\n=== {name} matrix ===")
    print("A =\n",np.round(A,4))
    print("\nCustom L =\n",np.round(Lc,4))
    print("\nNumPy L =\n",np.round(Ln,4))
    print("\nSciPy L =\n",np.round(Ls,4))
    print(f"\nErrors:  custom {err_c:.2e}, NumPy {err_n:.2e}, SciPy {err_s:.2e}")
```

```
 [0.6 0.8]]

SciPy L =
 [[1.  0. ]
 [0.6 0.8]]

Errors:  custom 0.00e+00, NumPy 0.00e+00, SciPy 0.00e+00

=== 3x3 matrix ===
A =
 [[ 2.7826 -1.6261  1.067 ]
 [-1.6261  3.3452  1.1337]
 [ 1.067   1.1337  3.1753]]

Custom L =
 [[ 1.6681  0.      0.    ]
 [-0.9748  1.5475  0.    ]
 [ 0.6396  1.1355  1.2152]]

NumPy L =
 [[ 1.6681  0.      0.    ]
 [-0.9748  1.5475  0.    ]
 [ 0.6396  1.1355  1.2152]]

SciPy L =
 [[ 1.6681  0.      0.    ]
 [-0.9748  1.5475  0.    ]
 [ 0.6396  1.1355  1.2152]]

Errors:  custom 5.44e-16, NumPy 4.44e-16, SciPy 4.44e-16

=== 5x5 matrix ===
A =
 [[ 3.8588  0.1508 -0.5381  0.8013  0.2812]
 [ 0.1508  1.6157 -0.6248 -1.2681 -1.2054]
 [-0.5381 -0.6248  6.9771  4.5739  1.9894]
 [ 0.8013 -1.2681  4.5739 10.5996  5.5925]
 [ 0.2812 -1.2054  1.9894  5.5925  4.8889]]

Custom L =
 [[ 1.9644  0.      0.      0.      0.    ]
 [ 0.0767  1.2688  0.      0.      0.    ]
 [-0.2739 -0.4759  2.5837  0.      0.    ]
 [ 0.4079 -1.0241  1.6249  2.597   0.    ]
 [ 0.1431 -0.9588  0.6086  1.3721  1.3024]]

NumPy L =
 [[ 1.9644  0.      0.      0.      0.    ]
 [ 0.0767  1.2688  0.      0.      0.    ]
 [-0.2739 -0.4759  2.5837  0.      0.    ]
 [ 0.4079 -1.0241  1.6249  2.597   0.    ]
 [ 0.1431 -0.9588  0.6086  1.3721  1.3024]]

SciPy L =
 [[ 1.9644  0.      0.      0.      0.    ]
 [ 0.0767  1.2688  0.      0.      0.    ]
 [-0.2739 -0.4759  2.5837  0.      0.    ]
 [ 0.4079 -1.0241  1.6249  2.597   0.    ]
 [ 0.1431 -0.9588  0.6086  1.3721  1.3024]]
```

Ex. 8 (d)

Suppose we want to simulate correlated random variables.

Let $R \in \mathbb{R}^{p\times p}$ be a correlation matrix.
Perform the **Cholesky factorization**:

$$R = LL^\top,$$

where L is lower triangular with positive diagonal entries.

Generate independent samples

Let $Z \in \mathbb{R}^{n\times p}$ be a matrix of independent standardized samples:

$$\mathbb{E}[Z] = 0, \quad \mathrm{Cov}(Z) = I_p$$

That is, each row z_i is an independent draw with identity covariance. Transform with L
Define the correlated samples:

$$X = ZL^\top.$$

So each row is $x_i = z_i L^\top$.

$$\mathrm{Cov}(x_i) = \mathbb{E}[x_i x_i^\top] = \mathbb{E}[\,(z_i L^\top)(L z_i^\top)\,] = L\,\mathbb{E}[z_i z_i^\top]\,L^\top = L I_p L^\top = LL^\top = R$$

Therefore, the transformed samples X have **exactly** the correlation matrix R. This proves that multiplying independent random draws by the Cholesky factor generates correlated variables with the desired correlation structure.

Ex. 8 (e)

We want to generate synthetic data that preserves **pairwise correlations** of the original dataset without using Cholesky (which may fail under multicollinearity).

Given $X \in \mathbb{R}^{n\times p}$:

- Sample mean: $\mu \in \mathbb{R}^p$
- Standard deviations: $s \in \mathbb{R}^p$
- Diagonal matrix: $D = \mathrm{diag}(s)$

Then the covariance is $S = \dfrac{1}{n-1}(X - \mathbf{1}\mu^\top)^\top(X - \mathbf{1}\mu^\top)$

and the correlation matrix is $R = D^{-1}SD^{-1}$

Eigen-decompose

$R = Q\Lambda Q^\top$

with $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_p)$.
Clip eigenvalues: $\tilde{\lambda}_i = \max(\lambda_i, \varepsilon), \quad \varepsilon > 0.$

Then

$R^{1/2} = Q\tilde{\Lambda}^{1/2}Q^\top,$

$\tilde{\Lambda}^{1/2} = \mathrm{diag}(\sqrt{\tilde{\lambda}_1}, \ldots, \sqrt{\tilde{\lambda}_p}).$

Generate latent samples.

Draw $Z \in \mathbb{R}^{m\times p}$ with rows $z_i \sim \mathcal{N}(0, I_p)$.
Form

$Y_{\text{lat}} = Z(R^{1/2})^\top.$

Then

$\text{Cov}(y_i) = R^{1/2} I_p (R^{1/2})^\top = R,$

so Y_{lat} has the desired correlation.

Marginal preservation (Gaussian copula).

For each column j:

- Rank the latent values $y_{1j}, \dots, y_{mj}$ into r_{ij}.
- Compute pseudo-uniforms

$u_{ij} = \dfrac{r_{ij} - 0.5}{m}.$

- Map through the empirical quantile of the original column $X_{\cdot j}$:

$y_{ij}^{(\text{marg})} = \hat{F}_j^{-1}(u_{ij}).$

This yields synthetic variables with the **original marginal distributions** and approximately the same correlations.

Since Z has $\mathbb{E}[Z] = 0$ and $\text{Cov}(Z) = I_p$,

$\text{Cov}\big(Z(R^{1/2})^\top\big) = R^{1/2}\,\text{Cov}(Z)\,(R^{1/2})^\top = R.$

Clipping eigenvalues ensures numerical stability even if R is nearly singular due to multicollinearity.

Ex. 9 (a)

Consider the quadratic

$$ax^2 + bx + c = 0,$$

where a, b, c are i.i.d. $\text{Uniform}(0, 1]$.

A quadratic has real roots if its discriminant is nonnegative:

$$\Delta = b^2 - 4ac \geq 0\ .$$

Ex. 9 (b)

The condition is $b^2 \geq 4ac$, i.e., $b \geq 2\sqrt{ac}$ (since $a, c, b \in (0, 1]$).

Because a and c are usually not tiny, the threshold $2\sqrt{ac}$ is often fairly large, leaving a relatively small slice of the $b \in (0, 1]$ range where the condition holds. We only have a chance when $ac \leq \frac{1}{4}$ (otherwise $2\sqrt{ac} > 1$). So the "eligible" region in (a, c) is limited, and within it b still needs to be big.

Putting that together, a reasonable ballpark guess is that the probability is around $1/4$; (≈ 0.25). Below $1/2$ and closer to a quarter than to a tenth.

Ex. 9 (c)

If we look closer at how a and c are distributed, their product ac is not "typically medium"; it is actually skewed strongly toward very small values (there are many more pairs with small product than with product near 1). Whenever ac is small, the threshold $2\sqrt{ac}$ is also small, so it's relatively easy for b to exceed it.

So, counter to the earlier intuition, the actual probability ends up being slightly larger than the rough guess of one-quarter.

Ex. 9 (d)

We want the probability

$P = \mathbb{P}(b^2 \geq 4ac),$

where $a, b, c \sim \text{Unif}(0, 1)$ i.i.d.

Integral formulation $P = \iiint_{[0,1]^3} \mathbf{1}\{b^2 \geq 4ac\}\, da\, db\, dc.$

For fixed (a, c), the contribution of b is

$\int_0^1 \mathbf{1}\{b \geq 2\sqrt{ac}\}\, db = \begin{cases} 1 - 2\sqrt{ac}, & ac \leq \frac{1}{4}, \\ 0, & ac > \frac{1}{4}. \end{cases}$

Double integral

Thus,

$P = \int_0^1 \int_0^{\min(1,\ \frac{1}{4a})} (1 - 2\sqrt{ac})\, dc\, da.$

Split into two regions:

$P = I_1 + I_2,$

where

- For $a \leq \frac{1}{4}$, upper limit = 1:

$I_1 = \int_0^{1/4} \int_0^1 (1 - 2\sqrt{ac})\, dc\, da.$

- For $a > \frac{1}{4}$, upper limit = $\frac{1}{4a}$:

$I_2 = \int_{1/4}^1 \int_0^{1/(4a)} (1 - 2\sqrt{ac})\, dc\, da.$

Evaluate I_1

$\int_0^1 (1 - 2\sqrt{ac})\, dc = 1 - 2\sqrt{a} \int_0^1 \sqrt{c}\, dc = 1 - \frac{4}{3}\sqrt{a}.$

So,

$I_1 = \int_0^{1/4} \left(1 - \frac{4}{3}\sqrt{a}\right)\, da = \left[a - \frac{8}{9}a^{3/2}\right]_0^{1/4} = \frac{5}{36}.$

Evaluate I_2

$\int_0^{1/(4a)} (1 - 2\sqrt{ac})\, dc = \left[c - \frac{4}{3}\sqrt{a}\, c^{3/2}\right]_0^{1/(4a)} = \frac{1}{12a}.$

So,

$I_2 = \int_{1/4}^1 \frac{1}{12a}\, da = \frac{1}{12}[\ln a]_{1/4}^1 = \frac{1}{6}\ln 2.$

$P = \frac{5}{36} + \frac{1}{6}\ln 2$

Numerical value:

$P \approx 0.2544.$

```
def analytic_probability():
    """
    Exact probability that b^2 >= 4ac
    when a, b, c ~ Uniform(0,1) i.i.d.
    """
    return 5/36 + (1/6)*math.log(2)

if __name__ == "__main__":
    result = analytic_probability()
    print(f"Exact probability = {result:.9f}")
    print("Formula: 5/36 + (1/6)*ln(2)")
```

```
Exact probability = 0.254413419
Formula: 5/36 + (1/6)*ln(2)
```

Ex. 9 (e)

The Monte Carlo simulation estimates the probability by randomly generating many triples (a, b, c) from the uniform distribution on $(0, 1)$. For each trial, we check whether the discriminant condition $b^2 \geq 4ac$ holds. Each success increases the count, and the ratio of successes to total trials gives the estimate of the probability.

As the number of runs N increases, the accuracy improves. With $N = 10{,}000$, the relative error is about 1.5%, but already by $N = 50{,}000$ it falls below 0.1%. For $N = 500{,}000$ and $N = 1{,}000{,}000$, the estimates are extremely close to the exact value $P = \frac{5}{36} + \frac{1}{6}\ln 2 \approx 0.2544$, with errors only on the order of 10^{-4}.

This behavior matches the theoretical expectation: the standard error of a Monte Carlo estimate decreases at the rate $1/\sqrt{N}$. In practice, this means that even a moderate number of runs produces a good approximation, and with one million runs the simulation essentially reproduces the analytic answer.

```
EXACT = 5/36 + (1/6)*math.log(2)

def mc(n, seed=42):
    rng = random.Random(seed)
    hits = sum((lambda a,b,c: b*b >= 4*a*c)(rng.random(), rng.random(), rng.random()) for _ in range(n))
    p = hits/n
    abs_err = abs(p-EXACT)
    rel_err = abs_err/EXACT
    se = (p*(1-p)/n)**0.5
    print(f"N={n:,}  Estimate={p:.6f}  AbsErr={abs_err:.6f}  RelErr={rel_err:.3%}  SE={se:.6f}")

for N in [10_000, 50_000, 500_000, 1_000_000]:
    mc(N)
```

```
N=10,000  Estimate=0.250500  AbsErr=0.003913  RelErr=1.538%  SE=0.004333
N=50,000  Estimate=0.254160  AbsErr=0.000253  RelErr=0.100%  SE=0.001947
N=500,000  Estimate=0.254766  AbsErr=0.000353  RelErr=0.139%  SE=0.000616
N=1,000,000  Estimate=0.254787  AbsErr=0.000374  RelErr=0.147%  SE=0.000436
```

Ex. 9 (f)

Whitening transforms the data so that it has mean 0 and covariance equal to the identity matrix.
In our case, the original variables $a, b, c \sim \mathrm{Unif}(0, 1)$ are already independent and have the same variance $1/12$.
The whitening procedure works correctly: after transformation, the mean is close to 0 and the covariance is close to I.

However, whitening also changes the distribution from uniform to something different.
Since the probability we want, $\mathbb{P}(b^2 \geq 4ac)$, is defined for the original uniform variables, whitening cannot be applied to improve the Monte Carlo estimation.

```
# Generate samples
N = 10000
X = np.random.rand(N, 3)   # a,b,c ~ Unif(0,1)

# Compute mean and covariance
mu = X.mean(axis=0)
Sigma = np.cov(X, rowvar=False)

# Whitening transform: (X - mu) * Sigma^{-1/2}
eigvals, eigvecs = np.linalg.eigh(Sigma)
Sigma_inv_sqrt = eigvecs @ np.diag(1/np.sqrt(eigvals)) @ eigvecs.T
X_white = (X - mu) @ Sigma_inv_sqrt

print("Original mean:", mu)
print("Original covariance:\n", Sigma)
print("\nAfter whitening:")
print("Mean (approx):", X_white.mean(axis=0))
print("Covariance (approx):\n", np.cov(X_white, rowvar=False))
```

```
Original mean: [0.497293   0.50133024 0.50164385]
Original covariance:
 [[ 0.08185983  0.00082704 -0.00073813]
 [ 0.00082704  0.08303602  0.00128633]
 [-0.00073813  0.00128633  0.08294829]]

After whitening:
Mean (approx): [-6.65800748e-17  4.12880841e-15 -3.67523789e-15]
Covariance (approx):
 [[ 1.00000000e+00 -9.73540902e-17 -6.16457469e-17]
 [-9.73540902e-17  1.00000000e+00 -4.68827453e-16]
 [-6.16457469e-17 -4.68827453e-16  1.00000000e+00]]
```

Ex. 10 (a)

Create a yield curve using constant maturity Yield We use the following tenors: 1M, 3M, 6M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 20Y, 30Y

```
ticker = ['DGS1MO', 'DGS3MO', 'DGS6MO','DGS1','DGS2', 'DGS3', 'DGS5','DGS7', 'DGS10','DGS20','DGS30']
sdt = dt.datetime(2014, 9, 28)
edt = dt.datetime(2024, 9, 27)
source = 'fred'
yieldcurve = pd.DataFrame(web.DataReader(ticker, source, sdt, edt))
yieldcurve = yieldcurve.dropna()
yieldcurve.to_csv('yieldcurvenona.csv')
```

```
yieldcurve.head(10)
```

DATE	DGS1MO	DGS3MO	DGS6MO	DGS1	DGS2	DGS3	DGS5	DGS7	DGS10	DGS20	DGS30
2014-09-29	0.01	0.02	0.05	0.11	0.58	1.06	1.77	2.21	2.50	2.95	3.18
2014-09-30	0.02	0.02	0.03	0.13	0.58	1.07	1.78	2.22	2.52	2.98	3.21
2014-10-01	0.01	0.02	0.04	0.10	0.53	1.00	1.69	2.12	2.42	2.87	3.12
2014-10-02	0.01	0.01	0.03	0.10	0.53	1.01	1.70	2.14	2.44	2.90	3.15
2014-10-03	0.01	0.01	0.03	0.11	0.57	1.05	1.73	2.16	2.45	2.89	3.13
2014-10-06	0.01	0.02	0.05	0.11	0.54	1.02	1.70	2.13	2.43	2.88	3.12
2014-10-07	0.02	0.02	0.04	0.10	0.52	0.98	1.64	2.06	2.36	2.81	3.06
2014-10-08	0.01	0.01	0.05	0.10	0.46	0.90	1.57	2.02	2.35	2.82	3.07

```
yieldcurve.tail(10)
```

DATE	DGS1MO	DGS3MO	DGS6MO	DGS1	DGS2	DGS3	DGS5	DGS7	DGS10	DGS20	DGS30
2014-10-10	0.02	0.01	0.04	0.10	0.45	0.89	1.55	1.99	2.31	2.77	3.03
2024-09-16	5.11	4.96	4.55	3.96	3.56	3.42	3.41	3.51	3.63	4.01	3.94
2024-09-17	5.05	4.95	4.55	3.99	3.59	3.45	3.44	3.53	3.65	4.02	3.96
2024-09-18	4.91	4.84	4.50	3.95	3.61	3.49	3.47	3.58	3.70	4.08	4.03
2024-09-19	4.89	4.80	4.46	3.93	3.59	3.47	3.49	3.60	3.73	4.11	4.06
2024-09-20	4.87	4.75	4.43	3.92	3.55	3.46	3.48	3.59	3.73	4.10	4.07
2024-09-23	4.85	4.72	4.40	3.91	3.57	3.47	3.51	3.62	3.75	4.12	4.09
2024-09-24	4.78	4.69	4.36	3.88	3.49	3.44	3.47	3.60	3.74	4.13	4.09
2024-09-25	4.79	4.69	4.36	3.89	3.53	3.49	3.52	3.65	3.79	4.18	4.14
2024-09-26	4.90	4.68	4.38	3.96	3.60	3.54	3.55	3.65	3.79	4.17	4.12
2024-09-27	4.90	4.68	4.35	3.90	3.55	3.49	3.50	3.60	3.75	4.15	4.10

Next steps: Generate code with `yieldcurve` New interactive sheet

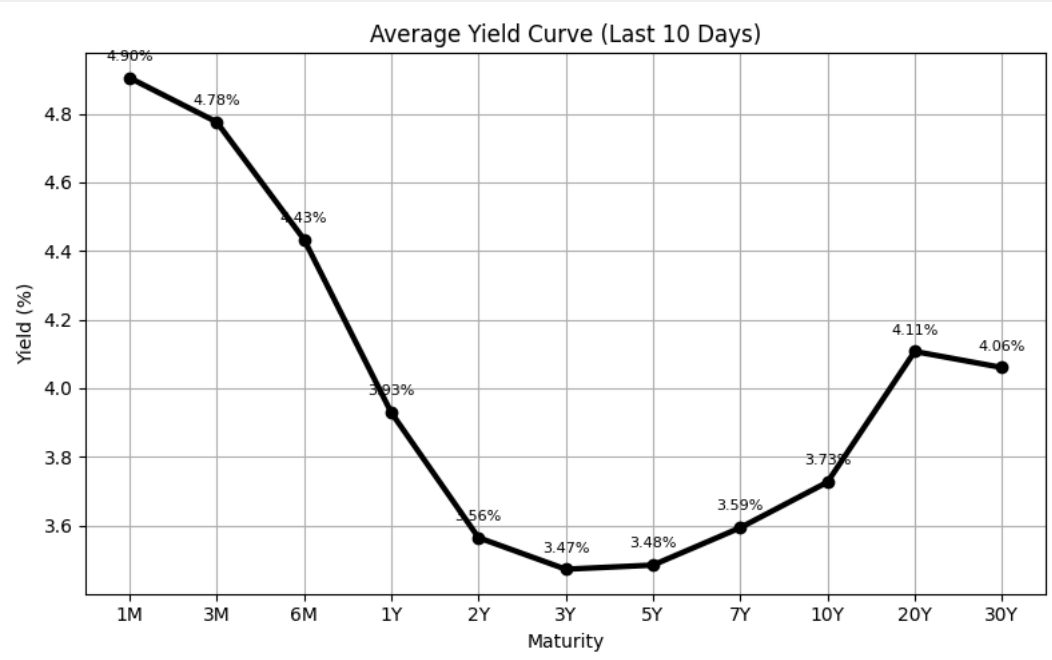
Ex. 10 (b)

US Treasury Yield Curve — Variable Definitions

Variable	Definition
DATE	Observation date (business days only, weekends/holidays excluded).
DGS1MO	Yield (%) on **1-Month US Treasury Bill** (constant maturity).
DGS3MO	Yield (%) on **3-Month US Treasury Bill** (constant maturity).
DGS6MO	Yield (%) on **6-Month US Treasury Bill** (constant maturity).
DGS1	Yield (%) on **1-Year US Treasury Note** (constant maturity).
DGS2	Yield (%) on **2-Year US Treasury Note** (constant maturity).
DGS3	Yield (%) on **3-Year US Treasury Note** (constant maturity).
DGS5	Yield (%) on **5-Year US Treasury Note** (constant maturity).
DGS7	Yield (%) on **7-Year US Treasury Note** (constant maturity).
DGS10	Yield (%) on **10-Year US Treasury Note** (constant maturity), often used as the benchmark risk-free rate.
DGS20	Yield (%) on **20-Year US Treasury Bond** (constant maturity).
DGS30	Yield (%) on **30-Year US Treasury Bond** (constant maturity), long-term government borrowing rate.

Ex. 10 (c)

```
maturity_labels = ["1M","3M","6M","1Y","2Y","3Y","5Y","7Y","10Y","20Y","30Y"]
cols = ["DGS1MO","DGS3MO","DGS6MO","DGS1","DGS2","DGS3",
        "DGS5","DGS7","DGS10","DGS20","DGS30"]

# Last 10 rows
df10 = yieldcurve.tail(10)

# Average curve
avg_curve = df10[cols].mean().values

# Plot average curve
plt.figure(figsize=(8,5))
plt.plot(maturity_labels, avg_curve, marker="o", linewidth=3, color="black")
plt.title("Average Yield Curve (Last 10 Days)")
plt.xlabel("Maturity"); plt.ylabel("Yield (%)")
plt.grid(True)

# Annotate each point with its yield value
for i, (x, y) in enumerate(zip(maturity_labels, avg_curve)):
    plt.text(i, y + 0.05, f"{y:.2f}%", ha="center", fontsize=8)

plt.tight_layout()
plt.show()
```




```
cols = ["DGS1MO","DGS3MO","DGS6MO","DGS1","DGS2","DGS3",
        "DGS5","DGS7","DGS10","DGS20","DGS30"]

# Last 10 rows
df10 = yieldcurve.tail(10)
```

```
# Average curve
avg_curve = df10[cols].mean().values

# Shape classifier
def classify_curve(y):
    diffs = np.diff(y)
    if np.all(diffs >= 0): return "upslope (normal)"
    if np.all(diffs <= 0): return "downslope (inverted)"
    if y[0] > y[-1] and 0 < np.argmin(y) < len(y)-1: return "inverted with hump/U-shape"
    return "mixed/flat"

shape = classify_curve(avg_curve)
print("Yield Curve Shape (last 10 days avg):", shape)

# Text interpretation
if "inverted" in shape:
    print("\nInterpretation: The curve is inverted at the short end and turns upward at long maturities, "
          "showing a humped/U-shape. This suggests markets expect slower growth and future Fed rate cuts, "
          "while long-term yields include a risk premium for inflation and fiscal uncertainty.")
elif "upslope" in shape:
    print("\nInterpretation: A normal upward-sloping curve, consistent with growth expectations and rising inflation.")
elif "downslope" in shape:
    print("\nInterpretation: Fully inverted, often a strong recession signal.")
else:
    print("\nInterpretation: Mixed/flat shape, signaling market uncertainty.")
```

```
Yield Curve Shape (last 10 days avg): inverted with hump/U-shape

Interpretation: The curve is inverted at the short end and turns upward at long maturities, showing a humped/U-shape. This suggests markets expect slower growth and future Fed ra
```

Ex. 10 (d)

```
# Calculate daily percentage change for each maturity
yieldcurve_pct = yieldcurve.pct_change() * 100  # convert to %

# Drop the first row (NaN because no previous day)
yieldcurve_pct = yieldcurve_pct.dropna()

# Show dimensions and a preview
print("Shape:", yieldcurve_pct.shape)
print(yieldcurve_pct.tail())
```

```
Shape: (2465, 11)
              DGS1MO    DGS3MO    DGS6MO      DGS1      DGS2      DGS3  \
DATE
2024-09-23 -0.410678 -0.631579 -0.677201 -0.255102  0.563380  0.289017
2024-09-24 -1.443299 -0.635593 -0.909091 -0.767263 -2.240896 -0.864553
2024-09-25  0.209205  0.000000  0.000000  0.257732  1.146132  1.453488
2024-09-26  2.296451 -0.213220  0.458716  1.799486  1.983003  1.432665
2024-09-27  0.000000  0.000000 -0.684932 -1.515152 -1.388889 -1.412429

                DGS5      DGS7     DGS10     DGS20     DGS30
DATE
2024-09-23  0.862069  0.835655  0.536193  0.487805  0.491400
2024-09-24 -1.139601 -0.552486 -0.266667  0.242718  0.000000
2024-09-25  1.440922  1.388889  1.336898  1.210654  1.222494
2024-09-26  0.852273  0.000000  0.000000 -0.239234 -0.483092
2024-09-27 -1.408451 -1.369863 -1.055409 -0.479616 -0.485437
```

Ex. 10 (e)

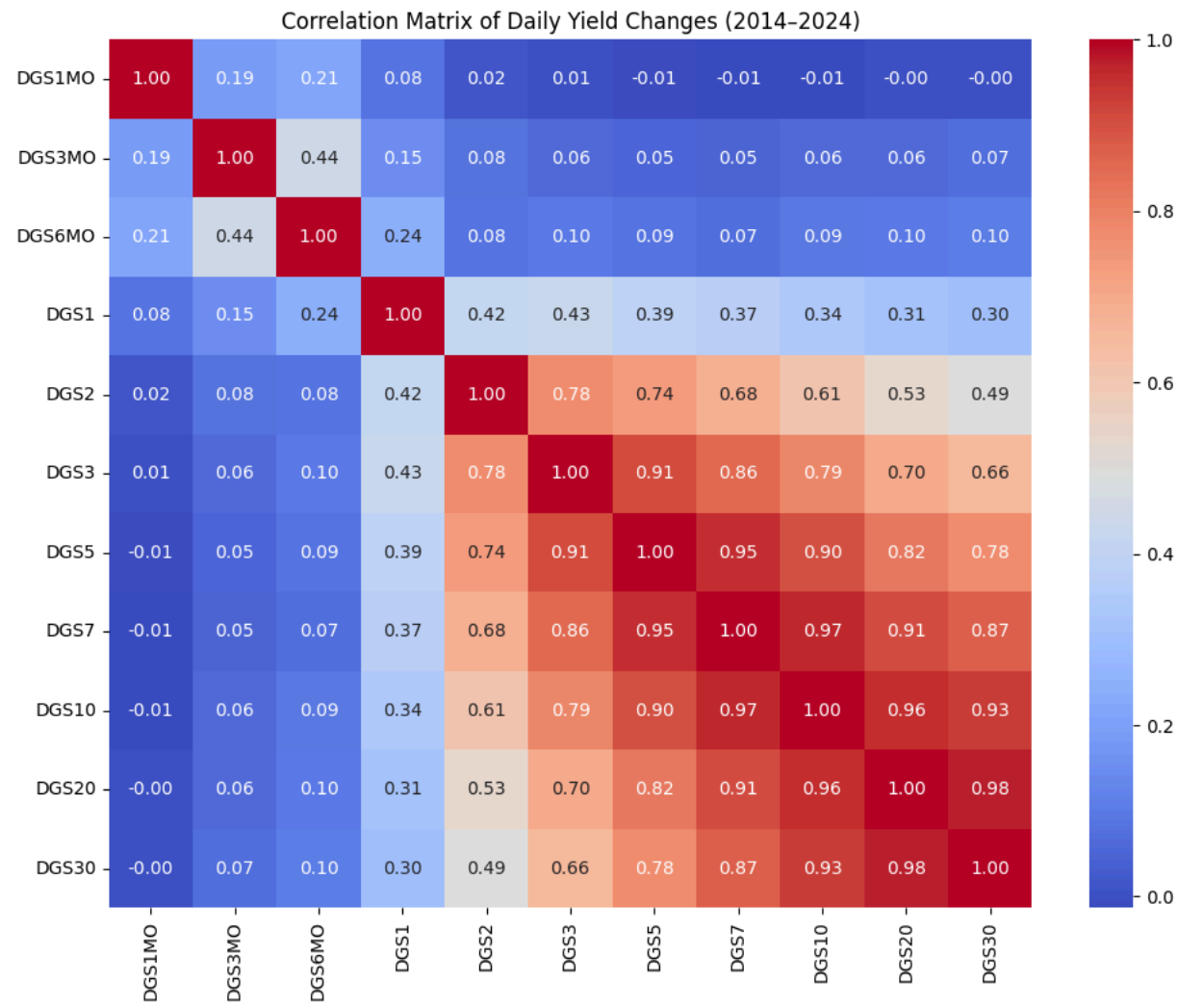
```
# Drop NaN for Short-term periods
yieldcurve_pct = yieldcurve_pct.replace([np.inf, -np.inf], np.nan).dropna()

# Mean and standard deviation for each maturity
stats = pd.DataFrame({
    "Mean % Change": yieldcurve_pct.mean(),
    "Std Dev % Change": yieldcurve_pct.std()
})
print(stats)
```

```
        Mean % Change  Std Dev % Change
DGS1MO       2.132561         29.659599
DGS3MO       1.822314         20.731896
DGS6MO       0.714407         12.679779
DGS1         0.330069          5.572245
DGS2         0.238056          5.088652
DGS3         0.177011          4.548485
DGS5         0.126032          3.887440
DGS7         0.088768          3.235702
DGS10        0.075093          3.022600
DGS20        0.050713          2.392571
DGS30        0.041344          2.167882
```

Ex. 10 (f)

```
# Correlation matrix of daily % changes
corr_matrix = yieldcurve_pct.corr()

print(corr_matrix)

# Heatmap for visualization
plt.figure(figsize=(10,8))
sns.heatmap(corr_matrix, annot=True, fmt=".2f", cmap="coolwarm", cbar=True)
plt.title("Correlation Matrix of Daily Yield Changes (2014–2024)")
plt.tight_layout()
plt.show()
```

```
          DGS1MO    DGS3MO    DGS6MO      DGS1      DGS2      DGS3      DGS5  \
DGS1MO  1.000000  0.187286  0.214612  0.082245  0.016076  0.005731 -0.009241
DGS3MO  0.187286  1.000000  0.443889  0.149841  0.079273  0.063138  0.050279
DGS6MO  0.214612  0.443889  1.000000  0.240770  0.082758  0.100727  0.086116
DGS1    0.082245  0.149841  0.240770  1.000000  0.415908  0.425434  0.385952
DGS2    0.016076  0.079273  0.082758  0.415908  1.000000  0.777310  0.741141
DGS3    0.005731  0.063138  0.100727  0.425434  0.777310  1.000000  0.910120
DGS5   -0.009241  0.050279  0.086116  0.385952  0.741141  0.910120  1.000000
DGS7   -0.011884  0.048313  0.073081  0.371681  0.682418  0.855127  0.953121
DGS10  -0.013219  0.058889  0.090964  0.344419  0.613000  0.788547  0.901484
DGS20  -0.004297  0.063351  0.099999  0.312424  0.529120  0.703019  0.821751
DGS30  -0.000421  0.069836  0.103654  0.303531  0.493448  0.657377  0.778036

            DGS7     DGS10     DGS20     DGS30
DGS1MO -0.011884 -0.013219 -0.004297 -0.000421
DGS3MO  0.048313  0.058889  0.063351  0.069836
DGS6MO  0.073081  0.090964  0.099999  0.103654
DGS1    0.371681  0.344419  0.312424  0.303531
DGS2    0.682418  0.613000  0.529120  0.493448
DGS3    0.855127  0.788547  0.703019  0.657377
DGS5    0.953121  0.901484  0.821751  0.778036
DGS7    1.000000  0.966178  0.906591  0.871989
DGS10   0.966178  1.000000  0.956957  0.934624
DGS20   0.906591  0.956957  1.000000  0.975053
DGS30   0.871989  0.934624  0.975053  1.000000
```



Ex. 10 (g)

Yes, the correlation matrix you calculated is positive semi-definite (since the minimum eigenvalue ≈ 0.017 > 0).

```
eigvals = np.linalg.eigvalsh(corr_matrix.values)   # corr_matrix from yieldcurve_pct.corr()
print("Minimum eigenvalue:", eigvals.min())
print("Is PSD (within tolerance)?", np.all(eigvals >= -1e-10))
```

```
Minimum eigenvalue: 0.017467877677062895
Is PSD (within tolerance)? True
```

Ex. 10 (h)

```
# Correlation matrix of yield changes
corr = yieldcurve_pct.corr()

# Eigen decomposition (symmetric matrix)
eigvals, eigvecs = np.linalg.eigh(corr.values)

# Sort in descending order
idx = np.argsort(eigvals)[::-1]
eigvals = eigvals[idx]
eigvecs = eigvecs[:, idx]

# Eigenvalues + explained variance
eigvals_df = pd.DataFrame({
    "Eigenvalue": eigvals,
    "Explained Variance %": eigvals / eigvals.sum() * 100
})

# Eigenvectors for all components
eigvecs_df = pd.DataFrame(eigvecs,
                          index=corr.columns,
                          columns=[f"PC{i+1}" for i in range(len(eigvals))])

# Results
print("Eigenvalues:\n", eigvals_df.round(4))
print("\nEigenvectors:\n", eigvecs_df.round(3))
```

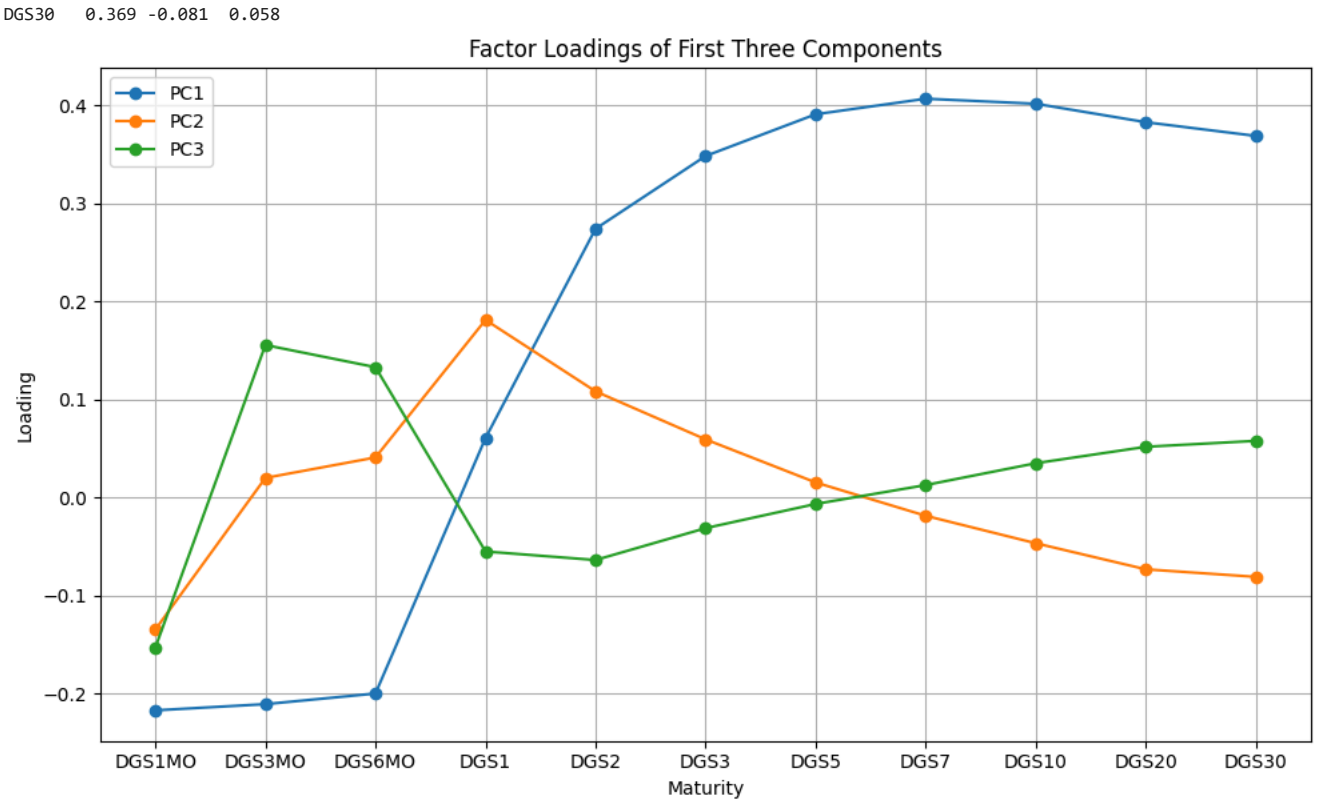
```
Eigenvalues:
     Eigenvalue  Explained Variance %
0       6.0494               54.9949
1       1.6352               14.8657
2       0.9091                8.2642
3       0.8594                7.8128
4       0.6300                5.7272
5       0.5271                4.7922
6       0.2270                2.0635
7       0.0895                0.8132
8       0.0346                0.3146
9       0.0212                0.1928
10      0.0175                0.1588

Eigenvectors:
```

```
            PC1    PC2    PC3    PC4    PC5    PC6    PC7    PC8    PC9   PC10  \
DGS1MO -0.006 -0.422  0.251 -0.870  0.019 -0.042  0.008  0.006 -0.002 -0.001
DGS3MO -0.042 -0.592  0.157  0.368  0.451 -0.531  0.041  0.005  0.002 -0.003
DGS6MO -0.060 -0.616  0.063  0.279 -0.252  0.687 -0.016  0.022 -0.011  0.004
DGS1   -0.190 -0.269 -0.646 -0.048 -0.566 -0.387  0.028  0.030  0.007 -0.007
DGS2   -0.306 -0.027 -0.428 -0.113  0.456  0.185 -0.682  0.041 -0.016 -0.010
DGS3   -0.363  0.011 -0.228 -0.071  0.263  0.155  0.498 -0.668 -0.156  0.015
DGS5   -0.388  0.052 -0.055 -0.034  0.155  0.099  0.354  0.421  0.682 -0.051
DGS7   -0.396  0.071  0.079 -0.007  0.023  0.006  0.173  0.420 -0.373  0.178
DGS10  -0.392  0.069  0.198  0.033 -0.089 -0.042 -0.001  0.219 -0.515  0.043
DGS20  -0.375  0.063  0.305  0.056 -0.203 -0.096 -0.200 -0.194  0.085 -0.772
DGS30  -0.363  0.057  0.338  0.067 -0.247 -0.126 -0.298 -0.333  0.314  0.607

         PC11
DGS1MO -0.004
DGS3MO  0.001
DGS6MO  0.011
DGS1   -0.006
DGS2   -0.003
DGS3   -0.000
DGS5   -0.197
DGS7    0.674
DGS10  -0.690
DGS20   0.169
DGS30   0.049
```

Ex. 10 (i)

```
# Correlation matrix of yield changes
corr = yieldcurve_pct.corr()

# PCA
pca = PCA()
pca.fit(corr)

# Factor loadings (eigenvectors scaled by sqrt(eigenvalues))
loadings = pca.components_.T * (pca.explained_variance_**0.5)

# Put loadings in DataFrame
loadings_df = pd.DataFrame(loadings[:, :3],
                           index=corr.columns,
                           columns=["PC1","PC2","PC3"])

print("Factor Loadings (first 3 components):\n")
print(loadings_df.round(3))

# Plot loadings
plt.figure(figsize=(10,6))
for i, comp in enumerate(loadings_df.columns, start=1):
    plt.plot(corr.columns, loadings_df[comp], marker="o", label=comp)
plt.title("Factor Loadings of First Three Components")
plt.xlabel("Maturity")
plt.ylabel("Loading")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```

```
Factor Loadings (first 3 components):

          PC1    PC2    PC3
DGS1MO -0.217 -0.135 -0.153
DGS3MO -0.211  0.020  0.156
DGS6MO -0.200  0.041  0.133
DGS1    0.061  0.181 -0.055
DGS2    0.274  0.108 -0.064
DGS3    0.349  0.059 -0.031
DGS5    0.391  0.015 -0.007
DGS7    0.407 -0.019  0.013
DGS10   0.402 -0.047  0.035
DGS20   0.383 -0.073  0.052
DGS30   0.369 -0.081  0.058
```



Ex. 10 (j)

In Principal Component Analysis (PCA), the **total variation** of the dataset is defined as the **sum of all eigenvalues** of the covariance (or correlation) matrix:

$$\text{Total Variation} = \sum_{i=1}^{n} \lambda_i$$

where λ_i are the eigenvalues.

Each eigenvalue represents the variance explained by one principal component (PC).

The explained variance ratio of the k-th component is:

$$\text{Explained Variance Ratio of PC}_k = \frac{\lambda_k}{\sum_{i=1}^{n} \lambda_i}$$

For a **correlation matrix** of n variables, the diagonal elements are all 1, so the trace is n.

Since the sum of eigenvalues equals the trace, the **total variation** is also n:

$$\sum_{i=1}^{n} \lambda_i = \text{trace}(\Sigma) = n$$

```
# Correlation matrix
corr = yieldcurve_pct.corr()

# Eigen decomposition
eigvals, eigvecs = np.linalg.eigh(corr.values)

# Sort descending
idx = np.argsort(eigvals)[::-1]
eigvals = eigvals[idx]

# Explained variance ratios
explained_var_ratio = eigvals / eigvals.sum()
cumulative_var_ratio = explained_var_ratio.cumsum()

# Put in DataFrame
explained_df = pd.DataFrame({
    "Eigenvalue": eigvals,
    "Explained %": explained_var_ratio * 100,
    "Cumulative %": cumulative_var_ratio * 100
})

print(explained_df.round(3))

# How many PCs to exceed 95%?
num_pcs = np.argmax(cumulative_var_ratio >= 0.95) + 1
print(f"\nNumber of PCs needed to explain >95% variation: {num_pcs}")
```

```
    Eigenvalue  Explained %  Cumulative %
0        6.049       54.995        54.995
1        1.635       14.866        69.861
2        0.909        8.264        78.125
3        0.859        7.813        85.938
4        0.630        5.727        91.665
5        0.527        4.792        96.457
6        0.227        2.064        98.521
7        0.089        0.813        99.334
8        0.035        0.315        99.648
9        0.021        0.193        99.841
10       0.017        0.159       100.000

Number of PCs needed to explain >95% variation: 6
```

Ex. 10 (k)

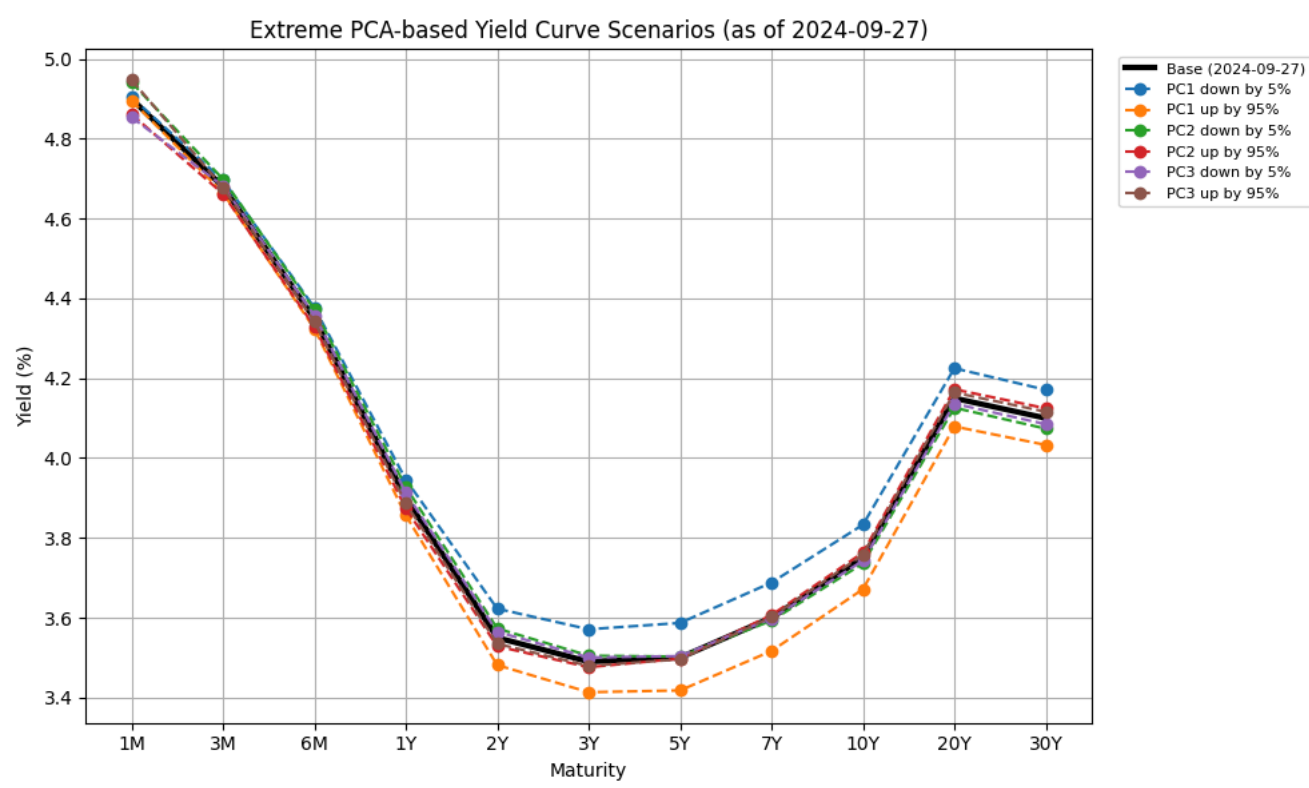
```
# Daily yield CHANGES in basis points (bp). (Yields are in %, so diff()*100 = bp.)
bp = yieldcurve[cols].diff().dropna() * 100.0     # shape T x 11

# PCA on the COVARIANCE of bp changes (loadings in bp per 1 unit factor move)
Sigma = np.cov(bp.values, rowvar=False)          # 11 x 11
eigvals, eigvecs = np.linalg.eigh(Sigma)         # symmetric
idx = np.argsort(eigvals)[::-1]                  # sort descending
eigvals = eigvals[idx]
eigvecs = eigvecs[:, idx]                        # columns are eigenvectors (loadings)

# Factor score time series (historical component moves)
scores = bp.values @ eigvecs                     # T x 11

# Extreme 5th/95th percentiles for the first 3 components
q05 = np.percentile(scores[:, :3], 5, axis=0)
q95 = np.percentile(scores[:, :3], 95, axis=0)

# Build 6 single-factor extreme scenarios: Δy(bp) = quantile * loading_vector
scenarios = {
    "PC1 down by 5%": q05[0] * eigvecs[:, 0],
    "PC1 up by 95%":  q95[0] * eigvecs[:, 0],
    "PC2 down by 5%": q05[1] * eigvecs[:, 1],
    "PC2 up by 95%":  q95[1] * eigvecs[:, 1],
    "PC3 down by 5%": q05[2] * eigvecs[:, 2],
    "PC3 up by 95%":  q95[2] * eigvecs[:, 2],
}
scen_bp = pd.DataFrame(scenarios, index=maturity_labels)
print("Extreme single-component scenarios (Δy in bp):")
print(scen_bp.round(1))

# BASE curve (latest): convert bp by /100
base_curve = yieldcurve[cols].iloc[-1].values    # in %
shocked_curves = {name: base_curve + (shock/100.0) for name, shock in scenarios.items()}
shocked_df = pd.DataFrame(shocked_curves, index=maturity_labels)
print("\nShocked yield curves (levels in %):")
print(shocked_df.round(3))
```

```
Extreme single-component scenarios (Δy in bp):
     PC1 down by 5%  PC1 up by 95%  PC2 down by 5%  PC2 up by 95%  \
1M              0.4           -0.4             4.1           -3.9
3M              1.6           -1.5             1.9           -1.8
6M              2.7           -2.6             2.2           -2.0
1Y              4.6           -4.4             2.8           -2.6
2Y              7.3           -6.9             2.3           -2.1
3Y              8.1           -7.7             1.5           -1.4
5Y              8.7           -8.2             0.3           -0.2
7Y              8.8           -8.3            -0.7            0.7
10Y             8.3           -7.8            -1.4            1.3
20Y             7.5           -7.1            -2.4            2.2
30Y             7.1           -6.7            -2.6            2.5

     PC3 down by 5%  PC3 up by 95%
1M             -4.6            4.8
3M              0.2           -0.2
6M              0.7           -0.8
1Y              1.3           -1.4
2Y              1.5           -1.5
3Y              1.0           -1.1
5Y              0.4           -0.4
7Y             -0.2            0.2
10Y            -0.7            0.8
20Y            -1.4            1.4
30Y            -1.5            1.6

Shocked yield curves (levels in %):
     PC1 down by 5%  PC1 up by 95%  PC2 down by 5%  PC2 up by 95%  \
1M            4.904          4.896           4.941          4.861
3M            4.696          4.665           4.699          4.662
6M            4.377          4.324           4.372          4.330
1Y            3.946          3.856           3.928          3.874
2Y            3.623          3.481           3.573          3.529
3Y            3.571          3.413           3.505          3.476
5Y            3.587          3.418           3.503          3.498
7Y            3.688          3.517           3.593          3.607
10Y           3.833          3.672           3.736          3.763
20Y           4.225          4.079           4.126          4.172
```

```
30Y           4.171          4.033           4.074          4.125

     PC3 down by 5%  PC3 up by 95%
1M            4.854          4.948
3M            4.682          4.678
6M            4.357          4.342
1Y            3.913          3.886
2Y            3.565          3.535
3Y            3.500          3.479
5Y            3.504          3.496
7Y            3.598          3.602
10Y           3.743          3.758
20Y           4.136          4.164
30Y           4.085          4.116
```

Ex. 10 (l)

```
#  Latest curve (Sep 27, 2024)
base_curve = yieldcurve.loc["2024-09-27", cols].values   # yields in %

# Build shocked curves (we already created scenarios in bp above)
shocked_curves = {name: base_curve + (shock/100.0) for name, shock in scenarios.items()}

# Plot
plt.figure(figsize=(10,6))
plt.plot(maturity_labels, base_curve, color="black", linewidth=3, label="Base (2024-09-27)")

for name, curve in shocked_curves.items():
    plt.plot(maturity_labels, curve, marker="o", linestyle="--", label=name)

plt.title("Extreme PCA-based Yield Curve Scenarios (as of 2024-09-27)")
plt.xlabel("Maturity")
plt.ylabel("Yield (%)")
plt.grid(True)
plt.legend(bbox_to_anchor=(1.02, 1), loc="upper left", fontsize=8)
plt.tight_layout()
plt.show()
```



Ex. 10 (m)

Both PCA and PAF extract the same underlying yield curve dynamics: Level, Slope, and Curvature, but they differ in focus. **PCA** explains the total variance (common plus idiosyncratic noise), while **PAF** explains only the common variance shared across maturities, filtering out noise; as a result, the shapes of the factors are nearly identical, but PAF gives slightly different magnitudes, making PCA more useful for statistical compression and hedging, and PAF better suited for interpreting underlying economic drivers.

```
# Correlation matrix of yield changes
corr = yieldcurve_pct[cols].corr()

# PCA (eigendecomp)
eigvals, eigvecs = np.linalg.eigh(corr)
idx = np.argsort(eigvals)[::-1]
eigvals, eigvecs = eigvals[idx], eigvecs[:, idx]
pca_loadings = eigvecs[:, :3] * np.sqrt(eigvals[:3])

# PAF (principal axis factoring) on raw yield changes
fa = FactorAnalyzer(n_factors=3, rotation=None, method="principal")
fa.fit(yieldcurve_pct[cols].dropna())   # pass data, not corr matrix
paf_loadings = fa.loadings_

# Results
print("PCA loadings (first 3):\n",
      pd.DataFrame(pca_loadings, index=maturity_labels).round(3))
print("\nPAF loadings (first 3):\n",
      pd.DataFrame(paf_loadings, index=maturity_labels).round(3))
```

```
PCA loadings (first 3):
          0      1      2
1M  -0.014 -0.539  0.239
3M  -0.104 -0.757  0.149
6M  -0.147 -0.787  0.060
1Y  -0.469 -0.344 -0.616
2Y  -0.753 -0.035 -0.408
3Y  -0.892  0.014 -0.217
5Y  -0.954  0.067 -0.053
7Y  -0.975  0.091  0.075
10Y -0.963  0.088  0.189
20Y -0.922  0.081  0.291
30Y -0.894  0.072  0.322

PAF loadings (first 3):
          0      1      2
1M   0.014  0.539  0.239
3M   0.104  0.757  0.149
6M   0.147  0.787  0.060
1Y   0.469  0.344 -0.616
2Y   0.753  0.035 -0.408
3Y   0.892 -0.014 -0.217
5Y   0.954 -0.067 -0.053
7Y   0.975 -0.091  0.075
```

```
10Y  0.963 -0.088  0.189
20Y  0.922 -0.081  0.291
30Y  0.894 -0.072  0.322
/usr/local/lib/python3.12/dist-packages/sklearn/utils/deprecation.py:151: FutureWarning: 'force_all_finite' was renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(
```

Ex. 11 (a)

A copula function is a multivariate uniform distribution over the $[0,1]$ hypercube. For a bivariate function $C(u,v):[0,1]\times[0,1]\to[0,1]$, we call $C(u,v)$ a bivariate copula function if it meets the following three conditions:

a. $C(0,u)=C(u,0)=0$

b. $C(1,u)=C(u,1)=u$

c. $C(u_2,v_2)-C(u_2,v_1)-C(u_1,v_2)+C(u_1,v_1)\geq 0$
for all $0\leq u_1\leq u_2\leq 1$ and $0\leq v_1\leq v_2\leq 1$
We want to show that $F(X)$, as a function of the random variable X, is a uniform distribution.
Since $F(x)$ is strictly increasing and continuous, it has an inverse F^{-1}. For any $u\in(0,1)$:

$$\mathbb{P}(F(X)\leq u)=\mathbb{P}\big(X\leq F^{-1}(u)\big)=F\big(F^{-1}(u)\big)=u$$

Thus, the distribution function of $F(X)$ is $H(u)=u$ on $[0,1]$, which is exactly the CDF of $\mathrm{Unif}(0,1)$.
Therefore,

$$F(X)\sim\mathrm{Unif}(0,1)$$

Ex. 11 (b)

We define

$$F(x_1,x_2)=C(F_1(x_1),F_2(x_2)),$$

where F_1 and F_2 are univariate CDFs, and C is a copula.

- Since $F_1(x_1)\in[0,1]$ and $F_2(x_2)\in[0,1]$, the input to C is always inside $[0,1]^2$.
  A copula C satisfies:
  1. **Groundedness:** $C(u,0)=C(0,v)=0$.
  2. **Margins:** $C(u,1)=u$, $C(1,v)=v$.
  3. **2-increasing property:** For any rectangle $[u_1,u_2]\times[v_1,v_2]\subset[0,1]^2$,
  $$V_C=C(u_2,v_2)-C(u_2,v_1)-C(u_1,v_2)+C(u_1,v_1)\geq 0$$
- By composition, $F(x_1,x_2)$ inherits these properties and is a valid **bivariate CDF**.
  The marginal of X_1 is defined as:
  $$F_{X_1}(x_1)=\lim_{x_2\to\infty}F(x_1,x_2)$$
  $$F_{X_1}(x_1)=\lim_{x_2\to\infty}C(F_1(x_1),F_2(x_2))$$
  Since $\lim_{x_2\to\infty}F_2(x_2)=1$, this becomes:
  $$F_{X_1}(x_1)=C(F_1(x_1),1)$$
  By the copula property $C(u,1)=u$, we get:
  $$F_{X_1}(x_1)=F_1(x_1)$$
  Similarly,
  $$F_{X_2}(x_2)=\lim_{x_1\to\infty}F(x_1,x_2)$$
  Compute explicitly:
  $$F_{X_2}(x_2)=\lim_{x_1\to\infty}C(F_1(x_1),F_2(x_2))$$
  Since $\lim_{x_1\to\infty}F_1(x_1)=1$, this becomes:
  $$F_{X_2}(x_2)=C(1,F_2(x_2))$$
  By the copula property $C(1,v)=v$, we get:
  $$F_{X_2}(x_2)=F_2(x_2)$$
- $F(x_1,x_2)=C(F_1(x_1),F_2(x_2))$ is a valid **bivariate distribution** (if C is a copula).
- Its **marginals** are exactly:
  $$F_{X_1}(x_1)=F_1(x_1),\quad F_{X_2}(x_2)=F_2(x_2)$$

Ex. 11 (c)

We verify the three copula properties.
Boundary conditions:

$$C(0,u)=\min(0,u)=0,\quad C(u,0)=\min(u,0)=0$$

and

$$C(1,u)=\min(1,u)=u,\quad C(u,1)=\min(u,1)=u$$

2-increasing property: For any $0\leq u_1\leq u_2\leq 1$ and $0\leq v_1\leq v_2\leq 1$,

$$\Delta=C(u_2,v_2)-C(u_2,v_1)-C(u_1,v_2)+C(u_1,v_1)$$

There are three possible cases:

- If $u_2\leq v_1$, then all terms equal the corresponding u's, and $\Delta=0$.
- If $v_2\leq u_1$, then all terms equal the v's, and again $\Delta=0$.
- If $u_1<v_1<u_2<v_2$, then the combination always yields $\Delta=0$.

Thus, $\Delta\geq 0$ in all cases.
Hence $C(u,v)=\min(u,v)$ satisfies all copula properties.

Ex. 11 (d)

We want to show that for any copula $C(u,v)$,

$$\max(u+v-1,0)\leq C(u,v)\leq\min(u,v)$$

Let

$$A=\{X_1\leq x_1\},\quad B=\{X_2\leq x_2\}$$

By definition of the copula,

$$C(u,v)=\Pr(U\leq u,V\leq v)=\Pr(A\cap B),$$

where

$$u=\Pr(A)=F_1(x_1),\quad v=\Pr(B)=F_2(x_2)$$

For any two events A and B,

$$\Pr(A)+\Pr(B)-1\ \leq\ \Pr(A\cap B)\ \leq\ \min(\Pr(A),\Pr(B))$$

Substituting $\Pr(A)=u$ and $\Pr(B)=v$ gives

$$u+v-1\ \leq\ C(u,v)\ \leq\ \min(u,v)$$

Since probabilities cannot be negative,

$$\Pr(A\cap B)\geq 0$$

Thus the actual lower bound is the maximum between 0 and $u+v-1$:

$$\max(u+v-1,0)\ \leq\ C(u,v)\ \leq\ \min(u,v)$$

For any bivariate copula function $C(u,v)$,

$$\max(u+v-1,0)\ \leq\ C(u,v)\ \leq\ \min(u,v).$$

Ex. 12

For any bivariate normal random vector (X, Y) and any function $h(x)$, prove that

$$\mathbb{E}[h(X)e^{-Y}] = \mathbb{E}[e^{-Y}]\,\mathbb{E}[h(X - \mathrm{Cov}(X, Y))]$$

Let (X, Y) be jointly Gaussian with mean vector $\mu = (\mu_X, \mu_Y)$ and covariance matrix

$$\Sigma = \begin{pmatrix} \mathrm{Var}(X) & \mathrm{Cov}(X, Y) \\ \mathrm{Cov}(X, Y) & \mathrm{Var}(Y) \end{pmatrix}$$

For any $t \in \mathbb{R}$, define the tilted expectation

$$\mathbb{E}_t[\cdot] = \frac{\mathbb{E}[(\cdot)e^{tY}]}{\mathbb{E}[e^{tY}]}$$

Under $\mathbb{E}_t$, the distribution of (X, Y) remains Gaussian with **shifted mean**

$$\mu_t = \mu + t\,\Sigma e_2,$$

where $e_2 = (0, 1)^\top$. Explicitly,

$$\mathbb{E}_t[X] = \mu_X + t\,\mathrm{Cov}(X, Y), \qquad \mathbb{E}_t[Y] = \mu_Y + t\,\mathrm{Var}(Y)$$

The covariance matrix is unchanged.

Therefore, under $\mathbb{E}_t$, the marginal of X is the same as the original X shifted by $t\,\mathrm{Cov}(X, Y)$. Hence for any function h:

$$\mathbb{E}_t[h(X)] = \mathbb{E}[h(X + t\,\mathrm{Cov}(X, Y))]$$

By definition,

$$\mathbb{E}[h(X)e^{tY}] = \mathbb{E}[e^{tY}]\,\mathbb{E}_t[h(X)] = \mathbb{E}[e^{tY}]\,\mathbb{E}[h(X + t\,\mathrm{Cov}(X, Y))]$$

Set $t = -1$

Finally, substitute $t = -1$:

$$\mathbb{E}[h(X)e^{-Y}] = \mathbb{E}[e^{-Y}]\,\mathbb{E}[h(X - \mathrm{Cov}(X, Y))]$$

The identity holds for any integrable function h, completing the proof.

Ex. 13 (a)

Suppose we have two random variables X_1 and X_2, and we define
$X_{1,2} = \min(X_1, X_2)$ and $X_{2,2} = \max(X_1, X_2)$.
Let $F_1(x) = \mathbb{P}(X_1 \le x)$ and $F_2(x) = \mathbb{P}(X_2 \le x)$.
If X_1 and X_2 are independent:

- Minimum:

$$F_{1,2}(x) = \mathbb{P}(X_{1,2} \le x) = 1 - \mathbb{P}(X_1 > x, X_2 > x) = 1 - (1 - F_1(x))(1 - F_2(x))$$

- Maximum: